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Seasonal Analysis of Customer Behavior Using RFM and K-means Clustering – A Case Study of Mobilegear

Master's thesis

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Anotace

Tato práce se zabývá analýzou zákaznického chování ve společnosti Mobilegear.cz pomocí RFM segmentace, K-means shlukování a analýzy sezónnosti. Hlavním cílem je identifikovat zákaznické segmenty, porovnat jejich chování a analyzovat vývoj zákaznické aktivity v čase.

Analýza vychází z transakčních dat a je provedena v rámci metodiky CRISP-DM. Hodnota zákazníků je měřena pomocí ukazatelů Recency, Frequency a Monetary a pro identifikaci skupin zákazníků je využito K-means shlukování. Sezónnost je analyzována na agregovaných časových datech.

Výsledky ukazují, že hodnota zákazníků není rovnoměrně rozložena a menší skupina zákazníků generuje vyšší podíl tržeb. Většina zákazníků se nachází ve středních segmentech, což naznačuje potenciál pro další rozvoj. Identifikované shluky zahrnují například nové, neaktivní, opakující se a hodnotné zákazníky. Sezónnost významně ovlivňuje zákaznickou aktivitu, zejména v obdobích jako Back-to-School a Vánoce, přičemž jednotlivé segmenty reagují odlišně.

Zjištění práce mají praktické využití v oblasti cíleného marketingu a řízení zásob.

Abstract

This thesis analyses customer behaviour in Mobilegear.cz using RFM segmentation, K-means clustering, and seasonality analysis. The main objective is to identify meaningful customer segments, compare their behaviour, and examine how customer activity changes over time.

The analysis is based on transactional data and follows the CRISP-DM framework. Customer value is measured using Recency, Frequency, and Monetary variables, and K-means clustering is applied to identify groups of customers with similar behaviour. Seasonality is analysed using aggregated time-based data.

The results show that customer value is unevenly distributed, with a relatively small group of customers generating a higher share of revenue. Most customers are concentrated in intermediate segments, indicating potential for further development. The clustering identifies distinct behavioural groups such as new, inactive, repeat, and high-value customers. Seasonality significantly influences customer activity, especially during periods such as Back-to-School and Christmas, with different segments reacting differently over time.

The findings provide practical implications for targeted marketing and inventory management.

Declaration

Prohlašuji, že jsem diplomovou práci na téma Sezónní analýza zákaznického chování pomocí RFM analýzy a K-means clusteringu – případová studie Mobilegear vypracoval samostatně pod vedením doc. Ahada Zareravasana, PhD a uvedl v ní všechny použité literární a jiné odborné zdroje v souladu s právními předpisy, vnitřními předpisy Masarykovy univerzity a vnitřními akty řízení Masarykovy univerzity a Ekonomicko-správní fakulty MU.

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Cíl práce, postup a použité metody:	**Cíl práce:** Cílem této diplomové práce je analyzovat sezónní trendy v nákupním chování zákazníků na trhu s použitými zařízeními Apple. Studie se zaměří na vliv vnějších faktorů, jako jsou uvedení nových produktů Apple na trh, sváteční období a zvýšená poptávka v období návratu do škol, a to vše na příkladu společnosti Mobilegear. Výsledkem práce bude návrh strategií pro optimalizaci řízení zásob, dynamické cenotvorby a cílených marketingových kampaní vycházejících ze sezónních vzorců. **Postup práce a použité metody:** Práce využije transakční data společnosti Mobilegear (včetně časových razítek nákupů, ID zákazníků, výše nákupů a typu produktu) a aplikuje metodu RFM analýzy a algoritmus K-means clustering. Na základě výsledků RFM analýzy budou zákazníci rozděleni do odlišných segmentů podle sezónních nákupních vzorců. Následně budou analyzovány prodejní trendy s cílem identifikovat období vrcholů a poklesů poptávky.
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Glossary

B2C	–	Business-to-Consumer
CLV	–	Customer Lifetime Value
CRM	–	Customer Relationship Management
CRISP-DM	–	Cross-Industry Standard Process for Data Mining
CV	–	Coefficient of Variation
DBSCAN	–	Density-Based Spatial Clustering of Applications with Noise
KPI	–	Key Performance Indicator
LTV	–	Lifetime Value
PCA	–	Principal Component Analysis
RFM	–	Recency, Frequency, Monetary
SSE	–	Sum of Squared Errors

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1 Introduction

Recent years have seen strong growth in e-commerce, which has also affected niche markets such as the sale of used and refurbished electronics. As competition increases, companies need to better understand customer behaviour and use data to support their decisions. In this context, customer segmentation plays an important role, as it allows firms to group customers based on their purchasing behaviour and design more targeted marketing activities, improve customer retention, and increase overall revenue (Tavakoli et al., 2018; Blattberg et al., 2008).

One of the most commonly used approaches to customer segmentation is the RFM (Recency, Frequency, Monetary) model. This method evaluates customers based on how recently they purchased, how often they purchase, and how much they spend. It is widely applied in retail because it provides a simple and practical way to describe customer behaviour and support marketing decisions (Chen et al., 2009; Blattberg et al., 2008). At the same time, the model has an important limitation, as it does not capture how customer behaviour changes over time.

This limitation becomes particularly relevant in the recommerce market, where demand is often influenced by seasonal factors such as holidays, back-to-school periods, or product launches. In such an environment, understanding only who the customers are is not sufficient; it is equally important to understand when they are most active. If seasonality is not considered, companies may face inefficient inventory planning, poorly timed marketing campaigns, and missed sales opportunities.

The problem addressed in this work arises from this gap. Mobilegear.cz, a Czech e-commerce company specialising in refurbished Apple devices, has access to detailed transactional data, but does not currently use a structured approach to customer segmentation that also reflects seasonal behaviour. As a result, decision-making may not fully capture how customer activity evolves over time.

To address this issue, RFM analysis is combined with K-means clustering and extended by adding a temporal perspective. Clustering makes it possible to identify groups of customers with similar behaviour (Han et al., 2023; Tavakoli et al., 2018), while the seasonal analysis helps to reveal how these groups behave across different periods. This combination provides a more complete view of customer behaviour than traditional static segmentation.

The objective is to analyse customer behaviour in the Mobilegear.cz dataset by linking segmentation with a time perspective. The analysis focuses on identifying customer segments, comparing their characteristics, and examining how their purchasing activity changes over time. Based on these findings, practical recommendations are proposed for marketing activities and inventory management.

The analysis is based on transactional data from Mobilegear.cz, including information on purchases, timestamps, customer identifiers, and product categories. Customer segments are identified using RFM analysis and K-means clustering, and seasonal patterns are subsequently analysed at the segment level.

The thesis is structured as follows. Chapter 2 presents the literature review, covering key concepts related to customer segmentation, RFM analysis, clustering, and seasonality. Chapter 3 describes the methodology, including research objectives, data, and analytical steps. Chapter 4 presents the analysis and findings. Chapter 5 discusses the results, outlines managerial implications, and concludes the thesis.

The contribution of this work lies in extending RFM-based segmentation with a temporal perspective and demonstrating how this approach can support practical decision-making. The results provide useful insights for Mobilegear.cz and can help improve marketing activities, demand planning, and overall business decisions.

2 Literature review

This chapter introduces the key concepts and analytical methods used in the thesis. Because the aim of the research is to analyse seasonal patterns in customer behaviour and to segment Mobilegear's customers using RFM analysis and K-means clustering, the literature review focuses on the frameworks that support these tasks. In addition to summarising existing research, the chapter also provides a synthesis of the literature, highlighting the main advantages and limitations of the selected approaches and explaining their relevance for the analytical design of this thesis.

The chapter first outlines the role of Customer Relationship Management (CRM) as the strategic basis for understanding customer value and guiding segmentation decisions. It then discusses customer segmentation, highlighting the shift toward behavioural and data-driven approaches that are particularly relevant in e-commerce and recommerce markets.

Next, the review presents the RFM analytic model, its logic, scoring procedures and extensions that address its limitations. The chapter also summarises key principles from data mining and the CRISP-DM framework, which structure the analytical workflow used to process large transactional datasets.

The final sections cover clustering methods, with emphasis on K-means, and the concept of seasonality in customer behaviour, including techniques for detecting seasonal patterns and integrating them into RFM-based segmentation.

Together, these theoretical foundations form the basis for the empirical analysis in the following chapters, where RFM, K-means and time-series methods are applied to Mobilegear's transactional data.

2.1 Customer Relationship Management

This section introduces Customer Relationship Management (CRM) as a strategic approach to managing long-term customer relationships and enhancing customer value. The chapter explains key CRM lifecycle activities, including acquisition, retention and development, and describes the main CRM dimensions: operational, analytical, and collaborative. It also highlights the growing importance of analytical CRM and data

mining in supporting segmentation, personalization, and decision making, particularly in retail and e-commerce.

Customer Relationship Management (CRM) has evolved from a simple tool for handling individual customer transactions into a broader strategic approach focused on building and sustaining long-term customer relationships. Modern CRM connects marketing activities, organizational processes, and information systems to better understand customer behaviour, enhance customer value over time, and allocate resources to the most profitable segments (Naim, 2021; Σιγάλα, 2018).

2.1.1 Concept and Evolution of CRM

CRM is commonly described as a managerial approach that translates relationship marketing principles into practice by identifying customer characteristics, supporting loyalty-building efforts, and improving customer retention (Naim, 2021). In this sense, CRM places customer relationships at the centre of business strategy and helps firms manage interactions in a structured and consistent way.

The concept extends beyond technology. CRM integrates strategic thinking, organizational procedures, and IT systems to gather and analyze customer information, enabling firms to manage interactions more effectively (Σιγάλα, 2018). A key part of this approach is its economic contribution, as CRM helps companies direct marketing budgets toward customers with the strongest potential for long-term profitability (Σιγάλα, 2018).

Historically, CRM systems were introduced mainly to improve the efficiency of customer transactions (Gurău, 2006). As marketing practices shifted away from transactional views and toward long-term relationship building, the role of CRM expanded to support customer-focused and process-oriented strategies (Gurău, 2006). Advances in digital technologies further accelerated this transition, allowing firms to target segments and individual customers with greater precision and to personalize communication and product offerings (Gurău, 2006).

This development paved the way for analytical CRM. The move from discrete transactions to ongoing customer interactions required the use of databases and analytical tools to support decision-making (Gurău, 2006). Data mining has become an essential component of this analytical layer because it helps firms identify patterns in customer data, improve relationship management, and create more effective customer strategies

(Chauhan, 2023). Insights generated from these analyses allow companies to design personalized marketing campaigns and product recommendations tailored to customer needs (Chauhan, 2023).

Ultimately, CRM aims to strengthen profitable long-term relationships by deepening understanding of customer preferences and behaviours (Kashani and Shahmirzaloo, 2017). Firms use CRM to identify potential customers, understand their needs and expectations, and enhance satisfaction and loyalty (Naim, 2021). In competitive markets, CRM also serves as a tool for differentiation by improving customer value and supporting efforts to retain high-potential customers (Σιγάλα, 2018; Naim, 2021). Through this targeted approach, CRM contributes to sustainable and profitable business growth.

2.1.2 CRM Processes and Dimensions

From a process perspective, CRM manages the entire customer lifecycle, typically structured around the stages of identification, acquisition, retention, and development. Many authors describe this lifecycle as beginning with the acquisition of new customers, followed by efforts aimed at retaining them and increasing their long-term value (Qin, 2023).

The acquisition phase focuses on attracting prospects who match the firm's target profile. Analytical techniques support this stage by helping organizations identify suitable potential customers and estimate their likely profitability (Thorleuchter et al., 2012). By applying these data-driven approaches, firms can target acquisition activities more precisely, which increases efficiency and improves conversion rates, which is an important benefit in business environments where acquisition is costly and traditionally yields low success. (Thorleuchter et al., 2012).

Once customers are acquired, CRM shifts its attention to retention and development. A commonly used framework highlights three core components of CRM: acquisition, retention, and customer expansion (Hasan et al., 2025). Retention becomes increasingly important as the costs associated with collecting and managing customer data continue to grow, pushing firms to design focused strategies for acquisition, retention, and re-engagement across different customer groups (Qin, 2023). The development stage aims to enhance customer value through actions that stimulate repeated purchases, encourage cross-buying, and increase overall consumption intensity (Qin, 2023), all of which contribute to higher customer lifetime value.

These lifecycle activities are supported by three principal CRM dimensions: operational, analytical, and collaborative. CRM encompasses a set of processes and technology tools designed to help organizations build long-term, profitable relationships with specific customers, relying on customer data and IT systems as a foundation (AZADMANESH, 2012). Operational CRM focuses on automating routine customer-facing processes, such as sales activities, service interactions, and marketing campaigns. Collaborative CRM facilitates information exchange and coordinated interactions across different touchpoints and departments to provide a seamless customer experience.

Analytical CRM plays a particularly important role in supporting data-driven decision making. Insights generated through analytical CRM help firms design more targeted marketing and promotional strategies and contribute to a comprehensive, 360-degree view of the customer (Xu and Walton, 2005). Its technological infrastructure typically combines CRM portals, data warehouses, predictive tools, and analytical engines that support tasks such as pattern detection, clustering, classification, and customer value evaluation (Xu and Walton, 2005). In this context, operational CRM focuses on process automation, while analytical CRM emphasizes the interpretation of customer characteristics and behaviour (Ngai et al., 2009).

Analytical CRM is also central to effective customer segmentation. Techniques such as clustering, classification, and customer value assessment allow firms to identify segments based on behaviour, profitability, or lifecycle stage and apply differentiated strategies to each customer tier (Xu and Walton, 2005; Qin, 2023). Through this approach, firms obtain a dynamic and continuously updated view of customer activity, which supports personalized communication, targeted marketing campaigns, and more effective resource allocation (Xu and Walton, 2005).

2.1.3 CRM in Retail and E-Commerce

Retailers, particularly in consumer electronics, rely on CRM to manage complex customer relationships and maximize lifetime value. Research shows that analytical CRM delivers the strongest performance benefits ($r = 0.52$), followed by collaborative ($r = 0.44$) and operational CRM ($r = 0.37$). This highlights the importance of insight generation and cross-functional coordination in extracting value from CRM initiatives (Rainy et al., 2024). Sector-level studies further indicate that combining CRM

with data-driven decision making is especially effective in retail, finance and healthcare, reflecting the relevance of CRM in data-intensive environments such as consumer electronics (Rainy et al., 2024).

These advantages depend on a solid analytical backbone. In retail settings, technologies such as CRM portals, data warehouses, predictive engines and pattern-identification tools enable firms to analyse purchasing activity, estimate customer value and design targeted offers (Xu and Walton, 2005).

E-commerce platforms extend these capabilities by generating rich digital footprints of customer behaviour. Many platforms include analytics features that reveal behavioural patterns, conversion trends and sales performance (Darshan and Jayabalan, 2024). At the same time, e-commerce applications automate key business processes, helping firms scale operations while reducing time and resource requirements (Darshan and Jayabalan, 2024). In this environment, the ability to extract timely, high-quality information becomes increasingly important, particularly as digital marketing is shaped by fragmented media channels and fast-evolving technologies such as mobile messaging and web-based supply chains (Ang and Buttle, 2006).

The analytical use of transactional data sits at the centre of these CRM applications. Analytical CRM refers to the interpretation of customer characteristics and behaviours to inform customer management strategies (Ngai et al., 2009). Insights derived from analytical CRM support the development of targeted marketing and promotional activities, while tools such as association rules, sequential pattern analysis, clustering and customer value assessment enable retailers to identify cross-selling opportunities, detect churn risks and evaluate high-value segments (Xu and Walton, 2005).

Loyalty programmes represent a widely used CRM tool in both retail and e-commerce. Their effectiveness depends on linking automated operational processes with analytics, as operational CRM manages routine interactions while analytical CRM interprets customer behaviour (Ngai et al., 2009). Customer management software plays a central role in this integration by consolidating interaction data, managing reward mechanisms and enabling personalized communication (Ang and Buttle, 2006).

Although most of the existing research focuses on primary retail and e-commerce, the same CRM principles are applicable to recommerce markets for refurbished consumer electronics. In these segments, analytical capabilities supported by technologies such as data warehouses

and predictive engines help recommerce firms understand customer preferences, anticipate demand for specific product categories, optimize pricing and develop targeted campaigns aimed at increasing customer lifetime value in a supply-constrained environment (Xu and Walton, 2005).

2.2 Customer Segmentation

This chapter examines customer segmentation as a core element of CRM and data-driven marketing. It explains how firms classify customers into meaningful groups based on value, behaviour, or characteristics, and how these segments guide acquisition, retention, and development strategies. The chapter outlines key segmentation objectives, including improving marketing efficiency, supporting resource allocation, and maximizing customer lifetime value. It also reviews major segmentation bases, including demographic, behavioural, RFM-based and CLV-based categories, and contrasts a-priori and post-hoc analytical approaches. Finally, the section discusses the role of big data, online CRM systems, and Customer Data Platforms in enabling dynamic and personalized segmentation in e-commerce and recommerce environments.

2.2.1 Concept, Objectives and Benefits of Customer Segmentation

Customer segmentation is closely linked to the evaluation of customer value. Value-based segmentation aims to identify profitable customers and develop strategies tailored to them (Ray and Mangaraj, 2016). Customer value is commonly defined through concepts such as lifetime value (LTV), customer lifetime value (CLV), customer equity and customer profitability (Ray and Mangaraj, 2016). These perspectives form the conceptual basis for segmentation methods that align marketing strategies with long-term financial outcomes.

Within the CRM context, segmentation is part of a broader customer management cycle. CRM is often described as a strategy that supports customer identification, attraction, retention and development, with segmentation acting as the first step that guides targeting, loyalty programmes and CLV-focused initiatives (Nie et al., 2020). As a result, segmentation is not a standalone analytical exercise but the entry point for designing personalized and value-oriented customer strategies.

The main purpose of segmentation is to increase the effectiveness and efficiency of marketing and resource deployment. Market segmentation and customer profiling are typically the initial phases of marketing planning (Kashani and Shahmirzaloo, 2017). By estimating CLV at the level of individual customers or segments, firms can make marketing initiatives more measurable and assess profitability more accurately, including in situations such as mergers or acquisitions (Hidalgo et al., 2008). Since a key goal of CRM is to maximize CLV to support long-term business performance, segmentation objectives are directly tied to sustainable value creation (Nie et al., 2020).

Segmentation provides both strategic and operational benefits. Financially, value-based segmentation requires that costs and benefits can be assigned correctly to each customer or segment, which then supports more precise resource allocation (Hidalgo et al., 2008). Strategically, accurate customer profiles enhance decision-making across multiple areas, from marketing to corporate valuation (Kashani and Shahmirzaloo, 2017). Recent advances also highlight the impact of AI-driven segmentation and personalization, as these techniques improve campaign effectiveness by tailoring offers to specific customer groups (Hemalatha, 2023). Together, these developments reinforce segmentation as a key pillar of data-driven CRM.

2.2.2 Bases and Methods of Customer Segmentation

Traditional segmentation approaches often rely on observable characteristics such as age, income or geographic location. These demographic and geographic methods provide a simple starting point, but they struggle to capture the behavioural diversity present within most customer bases. Segmentation can also be based on customer history with the firm, including RFM analysis, reflecting a shift from static attributes to behavioural patterns that better represent how customers interact with the company. (Rust and Verhoef, 2005).

Behavioural and psychographic approaches extend this view by examining engagement intensity, usage patterns and customer preferences. Grouping heterogeneous customers into meaningful segments helps firms design more effective marketing strategies (Duhita Wahyu Maulida and Dachyar, 2023). Behavioural segmentation in particular supports CRM by offering deeper insights into customer needs and decision-making processes, making it more actionable than segmentation

based solely on demographic characteristics (Duhita Wahyu Maulida and Dachyar, 2023).

Among data-driven methods, RFM (Recency, Frequency, Monetary) segmentation is widely used in CRM because it is simple to implement and effective in estimating customer value (Khajvand and Tarokh, 2011). By grouping customers based on recency of purchase, purchase frequency and spending levels, firms can prioritize retention activities, reactivation efforts and up-selling strategies (Khajvand and Tarokh, 2011). However, traditional RFM has limitations, particularly in its ability to differentiate between short-term and long-term relational patterns (Mahfuza et al., 2022).

These limitations have encouraged the development of enhanced RFM variants. Adding dimensions such as purchase volume improves the precision of customer segments (Mahfuza et al., 2022). Extended models like LRFMV aim to capture partnership characteristics and long-term relationship development, demonstrating how RFM-based segmentation continues to evolve to better represent relationship quality and potential (Mahfuza et al., 2022).

CLV-based segmentation advances this further by incorporating future value rather than relying solely on past behaviour. CLV represents the expected future cash flows generated by a customer within a CRM framework (Azadnia et al., 2011). It is used for evaluating customers, designing marketing strategies, recommending products and forming customer segments (Azadnia et al., 2011). Because CLV requires substantial transactional data, it has grown in relevance as firms increasingly recognise the importance of estimating long-term customer value in data-rich environments (Azadnia et al., 2011).

From a methodological perspective, segmentation approaches can be classified as a-priori or post-hoc. A-priori segmentation uses predefined rules, such as demographic criteria, that are set in advance and easy to apply but may not reflect underlying behavioural patterns (Rust and Verhoef, 2005). Post-hoc segmentation identifies groups empirically through data analysis. RFM and CLV-based approaches fall into this category, as they group customers into homogeneous segments based on observed behaviour (Khajvand and Tarokh, 2011). More sophisticated CRM models incorporate latent class or finite mixture techniques to capture unobserved heterogeneity, revealing segments that would not appear under a-priori methods (Rust and Verhoef, 2005).

2.3 RFM Analytic Model

Following chapter introduces the Recency–Frequency–Monetary (RFM) model as one of the most widely used behavioural segmentation tools in CRM. It explains the logic behind the three RFM dimensions, the scoring process, and the structure of the RFM grid used to classify customers into actionable value-based segments. The chapter highlights the strengths that make RFM popular, including its simplicity, low data requirements and strong predictive performance, while also addressing its limitations, such as restricted temporal awareness, limited behavioural granularity and challenges in long-term value estimation. Finally, it reviews major extensions of the classical model, including weighted, profitability-enhanced and engagement-augmented variants, as well as the use of RFM in advanced analytical and clustering workflows.

2.3.1 Definition and Logic of the RFM Model

The RFM model evaluates customers based on how recently they purchased (Recency), how frequently they buy (Frequency) and how much they spend (Monetary). It is widely recognised as an effective approach for segmenting consumer behaviour because it converts basic transactional information into practical indicators of customer value (Khumaidi et al., 2023). As a behaviour-based model, RFM assesses patterns in customer activity and uses them to generate forecasts, distinguishing it clearly from demographic segmentation (Khumaidi et al., 2023).

Its effectiveness lies in the combination of analytical strength and operational simplicity. Researchers often highlight RFM as one of the simplest and most powerful models for supporting CRM because it is easy to implement and requires only basic data inputs while still offering strong predictive accuracy (Yeh et al., 2009). Its predictive focus is central: RFM is designed to analyse ongoing behavioural signals and estimate future engagement based on these patterns (Yeh et al., 2009).

Logic of the Three Behavioural Dimensions

Recency measures the time since a customer's last purchase. Customers who purchased recently are typically more engaged and more likely to respond to marketing efforts. This dimension captures the most recent

purchase interval, which is strongly associated with near-term buying probability (Alie and Gustriansyah, 2024).

Frequency reflects how often a customer buys from the firm and serves as an indicator of behavioural loyalty. It describes spending intensity and distinguishes between infrequent purchasers and regular, high-value customers (Alie and Gustriansyah, 2024).

Monetary value captures the total amount spent by a customer, providing a direct financial assessment of their importance to the firm. This measure helps organizations prioritise customers according to their economic contribution (Alie and Gustriansyah, 2024).

Because these measures summarise actual behaviour over time, RFM performs well in predictive applications. It is often used to identify customers who may be at risk of churn, especially when recency or frequency begin to decline (Ibitoye et al., 2022). Many studies also describe RFM as a simple but effective foundation for market segmentation, since it can uncover meaningful differences in value across customer groups (Alie and Gustriansyah, 2024). Its continued use in retail, banking and telecommunications shows that it remains a reliable and practical tool within CRM (Yeh et al., 2009; Khumaidi et al., 2023).

2.3.2 RFM Scoring, RFM Grid and Customer Segments

RFM scoring transforms continuous transactional values into standardized scores that allow comparisons across customers. Its purpose is to translate behavioural patterns into numerical indicators that can help predict future activity (Miglautsch, 2000). The most common scoring method divides customers into quintiles and assigns a value from 1 to 5 for each RFM dimension. This approach supports consistent evaluation across large datasets and enables segmentation based on relative performance (Samidi et al., 2023).

A clear distinction between raw transaction data and the transformed ordinal scores is important, as the usefulness of RFM depends on how well the scoring process preserves meaningful behavioural differences (Samidi et al., 2023).

The RFM grid combines the three dimension scores into a three-digit code, such as 555, 431 or 112. This matrix provides a structured way to categorize customers and is widely used in strategic targeting. The grid enables the creation of RFM-based segments according to predefined criteria and scoring rules (Samidi et al., 2023). Its appeal lies in

its analytical clarity and simplicity, making it easy to implement across different business environments (Patron, 2004).

Common Customer Segments

Although segment definitions vary by industry, several categories consistently appear in empirical research:

- **Champions** – high-value customers who buy frequently, spend heavily and purchase recently. They often represent a small share of the customer base yet contribute disproportionately to revenue (Raharjo and Sugiati, 2025).
- **Loyal Customers** – regular purchasers who show sustained engagement.
- **Potential Loyalist**s – emerging high-value customers with strong future potential.
- **Promising / Recent Buyers** – recent purchasers with lower cumulative spend or frequency, representing opportunities for further development.
- **At-Risk Customers** – formerly valuable customers showing reduced recency or frequency.
- **Attention-Required Segments** – mid-tier groups with mixed or inconsistent behavioural patterns.

These segment categories allow firms to implement differentiated strategies, ranging from loyalty programmes for top-tier customers to retention initiatives for at-risk groups (Hermawan et al., 2024). Recent studies extend traditional RFM by incorporating additional variables or advanced analytical methods. For instance, combining RFM with demographic factors or clustering algorithms such as DBSCAN can increase segmentation precision and reveal more nuanced behavioural groups (Monalisa et al., 2023). These enhancements demonstrate how RFM can be integrated into broader machine-learning frameworks to support more sophisticated customer analysis.

2.3.3 Advantages and Limitations of Classical RFM

This subsection provides an overview of the main advantages and drawbacks of the classical RFM model. It summarises why RFM remains a widely used tool for behavioural segmentation, while also outlining the methodological limitations that motivate the development of extended or weighted RFM variants.

Strengths

- **Simplicity and interpretability:** The three dimensions, namely recency, frequency and monetary value, are easy for managers to understand, which helps encourage adoption (Yeh et al., 2009; Patron, 2004).

- **Minimal data requirements:** Only basic information such as purchase date, customer ID and transaction value is required, enabling fast implementation even with limited datasets (Kabasakal, 2020).

- **Strong actionability:** RFM segments naturally align with CRM objectives, supporting decisions such as targeting at-risk customers or identifying loyal buyers for upselling (Ibitoye et al., 2022; Hermawan et al., 2024).

- **Robustness and repeatability:** The method produces stable and homogeneous customer groups even in the absence of extensive supplementary data (Sarvari et al., 2016).

Limitations

- **Limited temporal awareness:** RFM captures only the most recent purchase and does not model purchase intervals, behavioural trends or seasonality (Miglautsch, 2000; Liao et al., 2024).

- **Lack of omnichannel integration:** All transactions are treated equally regardless of channel, which reduces accuracy in omnichannel settings (Tkachuk et al., 2025).

- **Coarse behavioural granularity:** Important behavioural factors such as product mix, returns or engagement metrics are excluded (Monalisa et al., 2023).

- **Arbitrary scoring mechanisms:** Equal weighting and quintile scoring may distort behavioural differences or mask important nuances (Chen, 2008; Sarvari et al., 2016).

- **Not a substitute for CLV:** RFM does not account for discounted future cash flows and therefore cannot fully estimate long-term customer value (Azadnia et al., 2011).

- **Transaction-only focus:** The model ignores non-purchase signals such as browsing activity or customer service interactions, which can be valuable indicators of engagement (Monalisa et al., 2023).

2.3.4 Extensions and Variants of RFM

The classical RFM model offers a simple way to describe customer behaviour, but several studies point out that its three basic dimensions cannot fully capture the variety of patterns seen in real data. These limitations have encouraged the development of extended versions that add behavioural, temporal or profitability-related variables. A common extension introduces Length (L), which reflects how long a customer has been active. Length helps distinguish stable long-term buyers from customers whose activity is concentrated in short periods (Marisa et al., 2019). Other authors add purchase volume (V), which helps separate frequent low-value buyers from occasional but high-value customers (Peker et al., 2017; Wang and Fu, 2025).

Digital businesses also expand RFM with engagement indicators such as browsing sessions, clicks or service interactions. These variables describe pre-purchase behaviour and often improve predictions related to churn or purchase propensity (Wang and Fu, 2025). Additional context variables, such as demographic or geographic characteristics, can make the resulting segments easier to interpret and align them with market realities (Peker et al., 2017).

Several extensions adjust the monetary dimension to better reflect profitability. Weighted RFM, margin-based monetary values or supervised versions that estimate optimal weights often outperform the equal-

weight structure of the classical model (Peker et al., 2017). Another stream of work replaces or enhances the monetary component with predicted customer lifetime value, making the model less dependent on historical spending alone (Peker et al., 2017; Azadnia et al., 2011).

Empirical research also highlights the value of integrating RFM into broader analytical workflows. Extended RFM vectors are widely used with clustering algorithms such as k-means, DBSCAN or fuzzy c-means to uncover nuanced behavioural groups (Ibitoye et al., 2022; Peker et al., 2017). Some firms embed RFM within customer-data platforms to capture cross-channel interactions, which is especially useful in omnichannel environments (Tkachuk et al., 2025). Studies also recommend validating the resulting segments through A/B testing or performance metrics to ensure that the enhanced models deliver practical improvements in marketing outcomes (Miglautsch, 2000).

Overall, the literature suggests that classical RFM is a solid baseline, while extended versions, including models incorporating tenure, engagement, channel-specific variables, weighting schemes or lifetime value, are more suitable when firms require greater precision, omnichannel coordination or long-term profitability analysis. (Wang and Fu, 2025; Sarvari et al., 2016; Chen, 2008).

2.4 Data Mining and CRISP-DM

This chapter introduces the role of data mining in behavioural segmentation and customer analytics. In this thesis, data mining refers mainly to the practical techniques used to prepare, analyse and interpret transactional data so that meaningful patterns in customer behaviour can be identified. These techniques help transform raw purchase records into information that supports segmentation, RFM analysis and the detection of seasonal trends that are relevant for recommerce markets.

The first part of the chapter explains how data mining contributes to understanding customer value and purchasing patterns, particularly in environments where demand fluctuates and transactions contain strong temporal signals. The second part presents the CRISP-DM framework, which provides a structured process for organising data mining projects. Its six phases offer a clear methodological foundation for tasks such as data preparation, modelling and evaluation, forming the basis for the analytical approach used later in the thesis.

2.4.1 Data Mining in Behavioural Segmentation and Recommerce

Data mining refers to the set of techniques used to extract useful patterns and insights from large datasets. In marketing and customer analytics, it helps firms understand how customers behave, how often they buy and which factors influence their decisions (Ngai et al., 2009). The core idea is simple: transform raw transactional data into information that supports better decisions. This process is especially relevant in digital and data-rich environments, where manual analysis would be slow and prone to error (Syadzali et al., 2020).

In the context of this thesis, data mining is closely tied to behavioural segmentation. Methods such as RFM analysis and clustering belong to the broader family of data mining tools because they summarise customer behaviour and reveal underlying patterns that cannot be observed directly. These techniques help group customers according to their value and purchasing habits, making it easier to target communication or adapt pricing strategies (Chen et al., 2011). Data mining also supports tasks such as preparing transaction records, extracting recency and frequency measures or creating features that capture seasonal activity.

Recommerce markets, including the sale of used Apple devices, are characterised by irregular supply and fluctuating demand, which makes the identification of behaviour patterns particularly important. Transactional data often contain seasonal purchase peaks and periods of slow activity, and data mining provides the tools needed to detect these trends and interpret them in a business context (Dhanushkodi et al., 2025). By uncovering such patterns, firms can better estimate customer value, anticipate periods of high demand and plan assortment and pricing strategies more effectively.

Overall, data mining acts as a practical bridge between raw data and actionable insight. It enables companies to analyse customer behaviour at scale, supports segmentation and helps identify seasonal trends that influence purchasing decisions. These principles form the basis for the structured analytical process introduced in the following section, where the CRISP-DM framework is presented as the methodological backbone of the empirical analysis (Ngai et al., 2009; Syadzali et al., 2020).

2.4.2 CRISP-DM Framework and Relevance for Segmentation

The Cross-Industry Standard Process for Data Mining (CRISP-DM) is one of the most widely adopted methodologies for structuring data mining projects. It provides systematic guidance for addressing key analytical challenges, including data preprocessing, model selection, interpretation and evaluation (Tripathi et al., 2021). Beyond offering procedural support, CRISP-DM acts as a methodological foundation that ensures data-driven initiatives remain aligned with business objectives and are implemented in a transparent and reproducible manner (Marchev and Marchev, 2025).

The framework is organized into six iterative phases:

1. Business Understanding

This phase translates business goals into data mining objectives. In customer analytics, it involves clarifying questions related to segmentation, retention or customer value, defining how RFM analysis or clustering should support CRM, and specifying relevant KPIs (Singgalen, 2023). CRISP-DM emphasizes that analytical models must be grounded in clearly defined business needs (Marchev and Marchev, 2025).

2. Data Understanding

The second phase focuses on collecting, describing and exploring data and evaluating data quality (Singgalen, 2023). For RFM and clustering projects, this includes examining transactional histories, behavioural indicators and temporal coverage, as well as identifying potential biases or gaps. Early identification of limitations is essential to setting realistic expectations for segmentation outcomes.

3. Data Preparation

Data preparation is typically the most labour-intensive phase, covering data cleaning, transformation and feature engineering (Tripathi et al., 2021). In RFM projects, this includes computing recency, frequency and monetary measures, deriving additional behavioural features and addressing missing values or outliers. For clustering, proper scaling,

normalization and variable selection are critical. CRISP-DM stresses that rigorous preparation is the foundation of reliable modelling.

4. Modelling

In this phase, analytical techniques are selected and applied to the prepared dataset. For segmentation, this includes defining RFM scoring rules and applying clustering algorithms such as k-means, hierarchical clustering or density-based methods (Tripathi et al., 2021). Model building typically involves parameter tuning and experimenting with alternative configurations to produce stable and interpretable segments.

5. Evaluation

The evaluation phase checks whether the developed models meet the defined business and technical objectives (Singgalen, 2023). In RFM and clustering, this includes assessing cluster quality, examining the interpretability of segment profiles and verifying that the segmentation aligns with business knowledge. This ensures that models are both statistically valid and practically useful for CRM and marketing.

6. Deployment

The final phase focuses on integrating analytical outputs into operational processes. In customer analytics, this involves implementing segment-based campaigns, automating scoring systems, scheduling regular segmentation updates and monitoring for model drift (Marchev and Marchev, 2025). Feedback gathered during implementation helps refine both the analytical models and associated business rules.

CRISP-DM is particularly valuable in big data environments where multiple data sources, high data volume and high velocity introduce significant integration and quality challenges. Its structured emphasis on data understanding and preparation allows firms to fully exploit big data while maintaining analytical rigour. For RFM and clustering applications in CRM, CRISP-DM provides a systematic roadmap that links business questions to operational segmentation, ensuring that customer analytics initiatives remain aligned with strategic goals and deliver sustainable value (Tripathi et al., 2021; Singgalen, 2023).

2.5 Clustering Methods

Clustering methods represent a key analytical foundation for data-driven customer segmentation because they reveal natural groupings within customer datasets without relying on predefined labels. By identifying latent behavioural patterns based on RFM attributes, purchase histories or other customer features, clustering provides a richer and more objective view of customer heterogeneity than traditional demographic or rule-based approaches.

This chapter reviews the principal clustering families used in marketing and CRM and explains their underlying logic, advantages and applications. It examines the K-means algorithm in detail, including its assumptions, distance measures and suitability for RFM-based segmentation, and discusses how to determine the appropriate number of clusters using internal validation indices. The chapter concludes by comparing K-means with alternative algorithms, outlining the conditions under which each method is most effective for building robust and interpretable customer segments.

2.5.1 Overview of Clustering in Customer Segmentation

In customer analytics, cluster analysis groups customers so that individuals within a cluster are highly similar while those in different clusters are as dissimilar as possible (Husein et al., 2021). This data-driven approach complements demographic or geographic segmentation by uncovering behavioural patterns that may not be apparent in advance.

Three main clustering families are commonly applied in marketing and CRM (Samidi et al., 2023):

1. Hierarchical Clustering

Hierarchical methods build a tree-like structure of nested clusters that allows analysts to explore segmentation at different levels of detail. Research demonstrates that hierarchical clustering can be combined with multi-criteria decision-making techniques to support business customer segmentation (Güçdemir and Selim, 2015). Agglomerative approaches begin with individual customers and merge them step by step,

while divisive methods start from a single large cluster and iteratively split it. The resulting dendrogram provides an intuitive visualization for exploring segment structures ranging from broad market divisions to smaller niche groups.

2. Partitioning Methods

Partitioning algorithms, such as K-means, divide the dataset into a predefined number of non-overlapping clusters. These methods are computationally efficient and produce clear segment assignments, which makes them attractive for operational CRM use. Studies combining K-means with RFM show that this approach can generate internally valid clusters, with strong performance indicators such as low Davies–Bouldin scores and high Silhouette values (Alzami et al., 2023).

3. Density-Based Methods

Density-based algorithms like DBSCAN identify clusters as dense regions in the feature space while treating low-density points as noise. Comparative studies show that density-based methods can capture irregularly shaped clusters and identify outliers more effectively than some alternative techniques (Türkmen, 2022). Other research highlights how DBSCAN can be integrated with K-means to create more comprehensive customer segments and inform detailed marketing strategies (Abednego et al., 2023).

Across all three families, clustering is often combined with RFM analysis. RFM scores provide a compact behavioural summarization, while clustering identifies groups of customers with similar RFM profiles in sectors such as online retail, banking and telecommunications (Samidi et al., 2023). Studies consistently show that RFM-based clustering produces clear, value-oriented segments that firms can target with differentiated marketing strategies (Hung and Dat, 2020; Chiang, 2012). More broadly, clustering is considered essential for partitioning heterogeneous customer populations into groups with similar needs, offering a foundation for product design, marketing strategy and CRM planning (Chan et al., 2012).

2.5.2 K-Means Algorithm: Principle, Assumptions and Distance Measures

K-means is one of the most widely used clustering algorithms in customer segmentation because it is simple, efficient and easy to interpret (Tabianan et al., 2022; Dewabharata, 2022). The algorithm partitions observations into k clusters by assigning each data point to the nearest centroid and then updating each centroid as the mean of its assigned members.

K-means minimises the within-cluster sum of squared distances, which corresponds to the sum of squared Euclidean distances between each point and its cluster centroid (Zheng et al., 2023). This optimisation objective drives the iterative process:

1. initialise *k* centroids,
2. assign each observation to the closest centroid,
3. recompute centroids as the mean of each cluster's members,
4. repeat until assignments stabilise (Kanungo et al., 2004; Dewabharata, 2022).

Because globally optimising this objective is NP-hard even when *k* equals 2, standard K-means typically converges to a local minimum (Jain and Obermayer, 2010; Zheng et al., 2023). For this reason, multiple random initialisations and improved seeding strategies such as k-means++ are often used to improve performance, along with heuristic variations of the algorithm (Mohamed Jafar and Sivakumar, 2014; Prasetyo et al., 2022).

In customer analytics, K-means is usually applied with Euclidean distance, which is consistent with its squared-error objective and well suited for continuous variables such as RFM measures (Rajput et al., 2015; Sari et al., 2022). Because Euclidean distance treats all dimensions equally, preprocessing steps such as scaling or normalisation are essential to prevent any single feature from dominating the clustering process (Prasetyo et al., 2022). Alternative distance measures and metaheuristic variants, including particle swarm optimisation or artificial bee colony–

based K-means, have also been proposed for specialised applications. (Jafar and Sivakumar, 2014; Raheem and Alabbas, 2022).

K-means relies on several assumptions. It performs best when clusters are roughly spherical, of similar size and variance, and when attributes are independent and approximately normally distributed (Tjhai et al., 2010). The method is sensitive to outliers, because centroids are means of assigned points, and to initial centroid placement, since it converges to local minima (Rajput et al., 2015; Sinaga and Yang, 2020). These characteristics require careful preprocessing, thoughtful selection of k and thorough robustness checks in customer segmentation projects.

Despite these limitations, K-means remains highly attractive for RFM-based segmentation. RFM attributes are continuous, low dimensional and intuitive to interpret, while K-means centroids correspond naturally to meaningful customer "prototypes" such as recent high spenders or inactive low-value customers. Empirical studies show that K-means applied to RFM variables can generate clear, actionable segments that firms use for inventory planning and targeted marketing (Dewabharata, 2022; Alzami et al., 2023).

2.5.3 Determining the Number of Clusters

Selecting the appropriate number of clusters (k) is a crucial step in partitioning and related clustering methods because it determines both the granularity and interpretability of the segmentation. Applied studies rarely depend on a single rule; instead, they typically combine multiple internal validation indices with visual diagnostics and then assess whether the candidate solutions make business sense (Ngai et al., 2009; Tripathi et al., 2021; Marchev and Marchev, 2025).

Several internal evaluation tools are widely used:

Elbow Method

The Elbow method evaluates the curve of within-cluster sum of squared errors (SSE) across different k values. The point where additional clusters yield only marginal reductions in SSE indicates a plausible range for k (Ngai et al., 2009; Tripathi et al., 2021). Because the method is heuristic and visually interpreted, it is primarily used to identify a search interval rather than to determine a single optimal value.

Silhouette Coefficient

The Silhouette coefficient compares how close each observation is to others in its own cluster (cohesion) relative to observations in the nearest neighbouring cluster (separation). The average silhouette score summarises overall clustering quality, and the k with the highest mean silhouette is typically preferred (Ngai et al., 2009; Tripathi et al., 2021). Low or negative values indicate potential misclassified or boundary points. Variants of the metric exist for categorical or fuzzy clustering (Marchev and Marchev, 2025).

Calinski–Harabasz (CH) Index

The CH index measures the ratio of between-cluster dispersion to within-cluster dispersion. Higher values signal cluster partitions with strong separation relative to compactness (Ngai et al., 2009; Tripathi et al., 2021). It is computationally efficient and frequently used alongside other indices.

Davies–Bouldin (DB) Index

The DB index evaluates the average similarity of each cluster to its most similar neighbour by combining intra-cluster scatter with inter-centroid distances. Lower values indicate better clustering performance (Ngai et al., 2009; Tripathi et al., 2021). Because the DB index can be sensitive to noise and initialization, it is often computed across multiple runs and random seeds.

Applied studies using RFM or transactional data commonly follow a multi-step workflow:

1. use the Elbow method to define a reasonable interval for *k*,

2. compute Silhouette, CH and DB values within this interval, typically across several K-means initialisations,

3. examine centroid profiles, cluster sizes and strategic interpretability before selecting the final number of segments (Ngai et al., 2009; Tripathi et al., 2021; Marchev and Marchev, 2025).

When indices disagree or when dataset characteristics violate assumptions of Euclidean distance, additional tools such as the gap statistic, information criteria or consensus-based approaches may be used.

Overall, best practice in customer segmentation involves combining quantitative metrics with managerial judgement. A valid clustering solution should show reasonable index values, reflect meaningful behavioural differences and result in segments that can be implemented operationally.

2.5.4 Strengths and Weaknesses of K-Means vs Alternative Methods

In customer analytics, K-means is often the default clustering technique, but alternative algorithms may be more suitable depending on the structure of the data and the segmentation objectives. Comparative considerations generally focus on cluster geometry, handling of noise or overlap, interpretability, sensitivity to parameters and scalability (Ngai et al., 2009; Dhanushkodi et al., 2025; Cirqueira et al., 2020; Samidi et al., 2023).

K-means minimises within-cluster squared error using Euclidean distance and produces clear, hard cluster assignments with straightforward centroid profiles. These characteristics make results easy to communicate to non-technical stakeholders (Ngai et al., 2009; Tripathi et al., 2021). It also scales efficiently to large datasets, which is essential in e-commerce and retail, and performs particularly well for low-dimensional, continuous variables such as RFM when appropriate scaling and transformations are applied (Chen et al., 2011; Singgalen, 2023). However, the method assumes clusters that are roughly spherical and of similar size, is sensitive to outliers and initialisation, and may fail to represent elongated or overlapping groups (Syadzali et al., 2020; Alzami et al., 2023; Hussein et al., 2024).

Hierarchical clustering offers a different perspective by building dendrograms that do not require specifying k in advance. This is useful

for organisations seeking multi-level segmentation structures, such as broad value tiers subdivided into smaller segments (Walsh et al., 2018; Çiçekli and Kabasakal, 2021). Various linkage methods, including single, complete, average or Ward, capture different notions of similarity, but results depend strongly on the chosen linkage. Classical hierarchical algorithms are computationally intensive, making them more appropriate for smaller datasets or aggregated features (Walsh et al., 2018; Cirqueira et al., 2020).

Density-based methods such as DBSCAN identify clusters as dense regions and classify low-density points as noise. These algorithms are suitable when clusters have irregular shapes or when outlier detection is important (Cirqueira et al., 2020; Huang, 2023). Although they do not require specifying *k*, they depend on parameters like eps and minPts, which must be carefully tuned. In customer analytics, density-based approaches are useful for isolating tight cohorts of repeat buyers and separating them from atypical or one-time customers (Anderson et al., 2007; Abednego et al., 2023).

Gaussian mixture models (GMMs) represent a model-based alternative in which data are assumed to come from a mixture of Gaussian components estimated via expectation–maximisation. GMMs support ellipsoidal, variable-sized clusters and produce probabilistic segment memberships, which can be valuable for uncertainty assessment and probabilistic targeting (Ngai et al., 2009; Dhanushkodi et al., 2025). However, they are more computationally demanding, sensitive to initialisation and model specification, and require information criteria or cross-validation to determine the number of components (Saxena et al., 2023; Dhanushkodi et al., 2025).

For RFM and transactional segmentation, the literature recommends a pragmatic workflow. First, compute interpretable RFM dimensions, transform skewed variables (e.g., log monetary, log frequency) and standardise features (Samidi et al., 2023; Marchev and Marchev, 2025). Second, use K-means as a fast and interpretable baseline with a carefully selected *k*, robust initialisation and internal validation (Tripathi et al., 2021; Singgalen, 2023). Third, if exploratory analysis suggests non-spherical or overlapping clusters, consider GMMs; if detecting outliers or irregular shapes is critical, apply DBSCAN or other density-based techniques (Cirqueira et al., 2020; Huang, 2023). Hierarchical clustering is suitable when multi-level segmentation or dendrogram visualisation is needed (Walsh et al., 2018; Çiçekli and Kabasakal, 2021).

Hybrid workflows are also common, for example using DBSCAN to identify and remove noise before applying K-means, or clustering K-means centroids hierarchically to create higher-level segment structures (Anderson et al., 2007; Menéndez et al., 2016; Samidi et al., 2023). Regardless of algorithm choice, studies emphasise evaluating candidate solutions using internal validation indices such as Silhouette, CH and DB, combined with stability checks and, importantly, the impact on downstream business outcomes such as campaign performance, churn rates or CLV before deploying segments in practice (Samidi et al., 2023; Singgalen, 2023; Cirqueira et al., 2020).

2.6 Seasonality in Customer Behaviour

Seasonality is a key element of customer behaviour and retail performance, as purchasing patterns often follow recurring temporal cycles shaped by weekly routines, monthly pay schedules or annual events. Understanding these rhythms is essential for accurate forecasting, inventory planning and the timing of marketing activities, especially in retail, e-commerce and recommerce environments. This chapter introduces the core components of time-series analysis, including trend, seasonality and irregular variation, and outlines how these elements influence sales dynamics, demand planning and managerial decisions in data-driven markets.

In addition to explaining methods for detecting and quantifying seasonality, the chapter reviews both classical and modern analytical techniques ranging from moving averages and seasonal indices to STL decomposition, ARIMA/SARIMA models and exponential smoothing. It also examines how seasonality can be integrated with RFM and clustering approaches through repeated scoring, rolling windows, time-weighted models and seasonal feature engineering. Together, these perspectives illustrate how incorporating seasonal dynamics leads to more accurate segmentation, improved prediction of customer activity and more informed decision-making in both e-commerce and recommerce settings.

2.6.1 Concept of Seasonality and Time Series in Marketing

Time-series analysis provides a fundamental framework for understanding how sales and customer behaviour change over time. In retail and e-commerce, key performance indicators, such as sales volume, revenue,

website traffic and campaign response, are inherently temporal, and their systematic variation shapes planning across finance, purchasing, marketing and logistics (Ramos et al., 2015). For this reason, marketing analytics commonly decomposes time series into trend, seasonal and irregular components.

Trend reflects the long-term direction of a series, whether growth, decline or structural change. Distinguishing trend from short-term fluctuations helps identify persistent shifts in demand, competitive dynamics or technological developments. Studies show that removing trend components can improve forecasting performance, particularly when combined with deseasonalisation (Zhang and Qi, 2005).

Seasonality refers to recurring patterns that repeat at fixed intervals. Some datasets exhibit seasonal behaviour defined as patterns that reappear at specific time intervals (Williams et al., 2023). In retail, these patterns typically involve weekly rhythms (weekday versus weekend), monthly cycles (e.g., payday effects) and annual fluctuations tied to holidays, weather or product release calendars. Ignoring such patterns can lead to incorrect interpretations of shocks or policy impacts (Gil-Alana et al., 2013). Seasonality also affects pricing and assortment decisions, as many product categories show predictable periods of high demand or discounting (Valpiani et al., 2015).

Irregular components capture residual variation after removing trend and seasonality. These deviations include both genuine shocks, such as unexpected campaigns or supply chain disruptions, and random noise. Forecasting models help distinguish expected patterns from anomalies by comparing predicted outcomes with realised data and flagging large residuals as potential exceptions (Williams et al., 2023).

Modern retail and e-commerce frequently involve multiple seasonalities within a single time series, such as daily, weekly and yearly cycles. Classical decomposition approaches like STL, X-12-ARIMA or X-13-ARIMA manage trend and single-seasonality patterns effectively (Bandara et al., 2021), but contemporary applications increasingly require methods capable of handling several seasonal frequencies simultaneously. For recommerce platforms specialising in used consumer electronics, these seasonalities may stem from weekly browsing behaviour, monthly pay cycles and annual upgrade waves, for example during new smartphone releases or the holiday season.

From a managerial perspective, the strategic relevance of seasonality lies in its influence on forecasting, inventory planning and marketing

timing. Inaccurate seasonal forecasts can lead to overstocking or stock-outs, both of which reduce profitability and harm customer satisfaction (Ramos et al., 2015). In recommerce, understanding seasonal demand for specific device types or conditions is particularly important for buy-back pricing, sourcing decisions and refurbishment capacity planning.

2.6.2 Methods to Identify and Measure Seasonality

A wide range of statistical techniques is available to detect and quantify seasonal patterns. In practice, analysts typically combine several complementary methods rather than rely on a single indicator, as each approach provides a different perspective on seasonal behaviour.

Descriptive Smoothing and Seasonal Indices

Moving averages are a simple but effective tool for smoothing short-term fluctuations and revealing underlying trend and seasonal components. For monthly data with annual seasonality, a 12-month moving average can be used to estimate the trend, which subsequently supports the calculation of seasonal indices. Classical multiplicative decomposition represents the observed series as:

$$Y_t = T_t \times S_t \times I_t$$

where T_t is the trend component, S_t the seasonal factor and I_t the irregular term (Ahmed, 2025). Seasonal indices provide an intuitive measure of how each month compares with an average or "typical" period, for example quantifying how much demand in December exceeds standard monthly levels.

Decomposition and Robust Techniques

Classical decomposition methods may struggle when seasonal patterns shift over time or when datasets contain outliers. More flexible techniques such as STL (Seasonal–Trend decomposition using Loess) iteratively estimate trend and seasonality via local regression and are better suited to non-stationary seasonal effects and noisy retail environments (Oum'bark et al., 2023). Such approaches are particularly helpful

when the strength or timing of seasonality changes due to market evolution or adjustments in promotional calendars.

Model-Based Approaches: ARIMA/SARIMA and Exponential Smoothing

Seasonality can also be incorporated directly into forecasting models. ARIMA combines autoregressive, differencing and moving-average components, while its seasonal extension, SARIMA, adds seasonal autoregressive and moving-average effects to capture periodicity (Du et al., 2025; Zhao, 2024). Exponential smoothing state-space models (ETS) offer another flexible option, representing level, trend and seasonality explicitly, with automated procedures that choose between additive or multiplicative forms. Comparative studies show that ARIMA, SARIMA and ETS often achieve strong forecasting accuracy, though performance varies across datasets and requires empirical validation (Yu et al., 2017; Zulu and Mwansa, 2022).

Across these methods, effective implementation depends on data quality (accurate timestamps, consistent intervals, treatment of missing values), the appropriate choice of seasonal frequency (e.g., weekly versus annual patterns) and the trade-off between model complexity and interpretability. For managerial use, seasonal indices and STL decompositions typically provide sufficient insight, while SARIMA or ETS models are preferred when accurate quantitative forecasts are needed for inventory or capacity planning.

2.6.3 Integrating Seasonality with RFM and Clustering

Classical RFM segmentation views each customer as a static point in a three-dimensional behavioural space, namely Recency, Frequency and Monetary value. This snapshot perspective overlooks temporal dynamics such as seasonal purchase peaks, replacement cycles and campaign-driven bursts, which are especially important in retail and recommerce (Ngai et al., 2009; Saxena et al., 2023; Syadzali et al., 2020). For this reason, recent research explores methods for making RFM and clustering models sensitive to seasonal effects.

One line of work uses seasonal windows or repeated scoring. In snapshot strategies, RFM scores are calculated at fixed intervals, such as

monthly or quarterly, and clustered independently at each time point. Analysts then examine how customers move between clusters across seasonal periods (Syadzali et al., 2020; Tripathi et al., 2021). Rolling-window approaches compute RFM on sliding horizons (e.g., the last 90, 180 or 365 days), allowing seasonal pulses to influence the scores. Window length becomes a design choice: shorter windows highlight seasonal signals but increase noise, while longer windows stabilise the scores but may smooth away meaningful seasonal variation. Many studies recommend aligning window choice with business rhythms such as campaign cycles or product life cycles (Ngai et al., 2009; Chen et al., 2011).

A second direction introduces time-weighted RFM. Instead of treating all purchases equally, more recent or seasonally relevant transactions receive higher weight. This can be implemented using exponential decay functions, explicit recency multipliers for key seasons or multi-criteria weighting methods such as fuzzy analytic network process (Dhanushkodi et al., 2025). Empirical evidence shows that weighted RFM often predicts future purchases and profitability better than the unweighted version, particularly when paired with clustering (Dhanushkodi et al., 2025; Syadzali et al., 2020).

A third approach enriches the RFM vector with time-series descriptors. Additional features such as seasonal purchase counts, seasonal spending totals or simplified RFM trajectories can be added to the segmentation dataset and analysed using multivariate or time-series clustering techniques (Ngai et al., 2009; Saxena et al., 2023; Chen et al., 2011). These approaches usually recover two orthogonal segmentation dimensions. The first dimension consists of value tiers (e.g., high/medium/low value or Gold/Silver/Bronze) based on aggregated or weighted RFM (Syadzali et al., 2020; Çiçekli and Kabasakal, 2021; Dhanushkodi et al., 2025). The second dimension reflects temporal patterns, distinguishing persistent buyers with steady purchasing, seasonal buyers concentrated around specific periods (such as Christmas), episodic or one-time customers and dormant or churn-prone shoppers (Saxena et al., 2023; Ngai et al., 2009). Combining value tiers with temporal cohorts yields hybrid labels, such as high-value persistent or medium-value seasonal, that are more informative for marketing and inventory decisions than pure RFM segmentation (Syadzali et al., 2020; Saxena et al., 2023).

From an operational perspective, season-aware clustering pipelines often keep K-means as the primary algorithm due to its scalability and interpretability, but apply it to seasonally enriched feature sets such as

weighted RFM, stacked seasonal RFM or simple seasonal summaries. More complex time-series clustering or LRFMS-style extensions are used selectively for priority customer groups where temporal shape is strategically important (Ngai et al., 2009; Tripathi et al., 2021).

Across all approaches, the literature stresses that season-aware segmentation must be validated not only with internal clustering metrics but also through downstream performance indicators such as repeat-purchase prediction, churn risk or profitability, as well as stability of clusters over time (Syadzali et al., 2020; Dhanushkodi et al., 2025; Chen et al., 2011). Where analytical resources are limited, a pragmatic strategy is to begin with K-means on standardised or weighted RFM combined with a small set of seasonal aggregates and then refine the approach for selected customer tiers. In recommerce markets, for example used Apple devices, this offers a feasible way to integrate seasonality into RFM-based segmentation while keeping the methodology operationally transparent and easy to implement.

2.7 Synthesis of the Literature and Methodological Implications

The literature reviewed in this chapter provides the theoretical background for the methods used in this thesis. However, the individual concepts need to be connected and interpreted in relation to the research objective. This section summarises the key insights and explains how they influence the methodological choices.

The literature highlights the importance of data-driven customer segmentation within Customer Relationship Management. Analytical CRM and data mining techniques allow firms to extract patterns from transactional data and use them for decision-making. Segmentation is therefore not only descriptive but also supports targeting, retention, and resource allocation (Ngai et al., 2009; Xu and Walton, 2005). This justifies the use of quantitative segmentation methods in this thesis.

RFM is consistently described as a simple and practical method for behavioural segmentation. It requires only basic transactional data and provides results that are easy to interpret, which makes it suitable for retail and e-commerce datasets (Yeh et al., 2009; Khumaidi et al., 2023). At the same time, the literature points out several limitations. RFM captures only a static snapshot of behaviour, does not reflect temporal

dynamics, and simplifies customer behaviour into a limited number of aggregated variables (Miglautsch, 2000; Liao et al., 2024). Despite these limitations, it remains a widely used baseline method. In this thesis, RFM is therefore used as the initial step to represent customer behaviour in a structured and interpretable way.

The scoring of RFM variables is typically based on quantiles. This approach divides customers into groups and assigns ordinal scores that allow comparison across the dataset. The literature identifies this method as the most common approach, mainly because it is easy to implement and works well for large datasets (Samidi et al., 2023).

At the same time, alternative approaches exist. Some studies propose weighted RFM models, where different importance is assigned to individual dimensions, or extended versions of RFM that incorporate additional variables such as customer lifetime value or engagement metrics (Peker et al., 2017; Azadnia et al., 2011). These approaches aim to capture customer behaviour more precisely and address limitations of the classical model.

However, these methods are more complex and often require additional data or model assumptions. In contrast, quantile-based scoring provides a simple and transparent way to transform transactional data into comparable indicators. Despite this, the literature also points out that quantile-based scoring can be somewhat arbitrary and may mask differences between customers (Chen, 2008; Sarvari et al., 2016). Based on this trade-off, quantile-based scoring is used in this thesis as a standard and widely accepted approach, while acknowledging that it simplifies the underlying behaviour.

To move beyond rule-based segmentation, clustering methods are used to identify groups of customers with similar characteristics. Among the available algorithms, K-means is frequently used because it is simple, computationally efficient, and produces results that are easy to interpret (Ngai et al., 2009; Tripathi et al., 2021). These properties are important in practical applications, where segmentation results need to be clearly understood and used in decision-making. K-means also works well with continuous variables such as RFM measures (Rajput et al., 2015).

Other clustering methods offer different advantages. Hierarchical clustering allows multi-level segmentation but is computationally more demanding, especially for larger datasets (Walsh et al., 2018). Density-based methods such as DBSCAN can identify irregular cluster shapes and detect noise, but require careful parameter tuning and are less

straightforward to interpret (Cirqueira et al., 2020; Huang, 2023). The literature therefore suggests that method selection should consider both analytical performance and practical usability. Based on this trade-off, K-means is selected in this thesis as a suitable method for customer segmentation.

The literature also emphasises that customer behaviour is not static and often follows seasonal patterns. Traditional RFM and clustering approaches do not capture these temporal effects (Ngai et al., 2009; Syadzali et al., 2020). To address this, different extensions are proposed, including repeated scoring, time-weighted models, and the integration of time-series features.

The literature distinguishes between simple descriptive approaches and more advanced statistical models for analysing seasonality. Descriptive methods, such as seasonal indices, provide an intuitive way to identify recurring patterns and are widely used in practice due to their simplicity and interpretability (Ahmed, 2025). More advanced approaches, including decomposition techniques or ARIMA-based models, allow more precise modelling of temporal dynamics, but require stronger assumptions and higher complexity (Du et al., 2025; Zhao, 2024).

In this thesis, a descriptive approach is used, as it provides sufficient insight into seasonal patterns while maintaining transparency and interpretability of the results.

Overall, the literature supports a combined approach. RFM provides a structured representation of customer behaviour, K-means identifies groups with similar characteristics, and seasonality analysis adds a temporal perspective. Each method has limitations, but together they form a consistent and practical framework for analysing customer behaviour in the given context.

3 Methodology

This chapter outlines the methodological approach used in the thesis to analyse customer behaviour in the Mobilegear.cz dataset. It defines the research objectives and questions and describes the framework, data, and analytical methods applied in the study.

The chapter begins by presenting the research objectives and research questions, which focus on customer segmentation and the analysis of seasonal patterns in purchasing behaviour. It then introduces the CRISP-DM framework, which provides a structured approach to organising the analytical process and links business objectives with data analysis.

This is followed by a description of the dataset used in the study, including its structure and key characteristics. The final section presents the analytical approach, explaining how transactional data are transformed into RFM metrics, how customer segments are identified using clustering techniques, and how temporal patterns in customer behaviour are analysed.

Together, these elements establish the methodological foundation of the thesis and provide a clear basis for the empirical analysis presented in the following chapter.

3.1 Research Objectives

The aim of this thesis is to analyse customer behaviour in the Mobilegear.cz dataset by combining customer segmentation with a seasonal perspective. The study builds on the concepts introduced in the previous chapter and applies them to real transactional data from a recommerce business.

The first objective is to develop a customer segmentation model using RFM analysis and K-means clustering. This approach groups customers based on their purchasing behaviour, specifically recency, frequency and monetary value. By applying clustering to RFM scores, the analysis aims to identify clear and meaningful customer segments.

The second objective is to describe and compare the identified customer segments. This includes analysing their key characteristics and differences in purchasing behaviour, in order to better understand how customer value is distributed across the dataset.

The third objective is to extend the analysis by adding a time perspective and examining how purchasing behaviour changes over time. The focus is on seasonal patterns, such as recurring changes in demand across different months or periods.

The final objective is to use the results of the analysis to provide practical recommendations. Based on the identified segments and their seasonal behaviour, the thesis aims to suggest how marketing activities and inventory decisions can be improved in the recommerce context.

To support these objectives, the following research questions are defined:

RQ1: How can RFM-based segmentation combined with K-means clustering be used to identify distinct customer segments in the Mobilegear.cz dataset?

RQ2: How does purchasing behaviour differ across the identified customer segments?

RQ3: How does purchasing behaviour of different customer segments vary over time, particularly with respect to seasonal patterns?

RQ4: How can insights from customer segmentation and seasonal behaviour be used to support marketing and inventory decisions?

3.2 Framework – CRISP-DM

This thesis follows the CRISP-DM framework as a structured approach to the analysis of customer behaviour. The framework provides a clear sequence of steps that connect business objectives with data analysis and interpretation. In this study, CRISP-DM is not applied as a strict linear process, but rather as a guiding structure for organising the analytical workflow.

In the business understanding phase, the main objective is to analyse customer behaviour and identify meaningful customer segments in the Mobilegear.cz dataset. The focus is on understanding how customers differ in terms of purchasing behaviour and how these differences can be

used to support marketing and inventory decisions. A particular emphasis is placed on identifying seasonal patterns in customer activity.

The data understanding phase focuses on exploring the available transactional data. The dataset consists of order-level and item-level data, which are linked through a common order identifier. This phase includes examining the structure of the data, time coverage, key variables and potential limitations, such as missing values or inconsistencies between tables.

In the data preparation phase, the raw transactional data are transformed into a format suitable for analysis. This includes joining datasets, filtering relevant transactions and constructing customer-level variables. The key step is the calculation of RFM metrics, which summarise customer behaviour in terms of recency, frequency and monetary value. Additional transformations, such as aggregation over time, are applied to support the seasonal analysis.

The modelling phase applies RFM scoring and K-means clustering to identify customer segments. Customers are first assigned RFM scores based on their relative position in the dataset. These variables, after standardisation, are then used as input for clustering, which groups customers into segments with similar behavioural patterns.

In the evaluation phase, the resulting customer segments are assessed in terms of interpretability and business relevance. The analysis focuses on whether the identified segments represent meaningful differences in customer behaviour and whether they can be used to support practical decision-making. Seasonal patterns are also evaluated at the segment level.

The final phase, deployment, is not implemented in a technical sense, but is addressed through the interpretation of results. The findings are translated into practical recommendations for marketing and inventory management in the context of the Mobilegear.cz business.

3.3 Data

The analysis in this thesis is based on transactional data obtained from the Mobilegear.cz company. The dataset was provided by the company's IT department and contains records of customer purchases in a recommerce environment focused on used Apple devices.

The data cover the period from 1 January 2023 to 1 November 2025 and include only B2C transactions. The dataset is structured into two main tables: an orders table and an item-level sales table. These tables are linked through a shared order identifier, which allows transactions to be analysed both at the order level and at the individual product level.

The orders table contains 31,247 records and includes information such as order identifier, timestamps, e-mail hash, total order value and payment date. The sales table contains 33,086 records and provides detailed information about individual items within each order, including product type, product name, price, currency and timestamp of the transaction.

This structure allows the analysis to capture both overall customer behaviour and detailed product-level characteristics. It also enables the construction of customer-level metrics, which are necessary for the RFM analysis.

3.4 Analytical Approach

This section builds on the CRISP-DM framework described above and provides a more detailed description of the analytical steps. The analytical approach combines customer segmentation techniques with a time-based analysis of purchasing behaviour. The process follows a sequence of steps that transform raw transactional data into interpretable customer segments and allow their behaviour to be analysed over time.

The first step involves the calculation of RFM metrics at the customer level. Transactional data are aggregated using the e-mail hash, and each customer is evaluated based on recency (time since the last purchase), frequency (number of transactions) and monetary value (total spending). These metrics provide a simplified representation of customer behaviour. In addition, outliers are detected and removed to improve the robustness of the clustering results.

In the next step, customers are assigned RFM scores using a quantile-based approach. Each of the three dimensions is divided into five groups, and customers receive a score from 1 to 5 based on their relative position in the dataset. This transformation makes the variables comparable and suitable for further analysis. Before clustering, the RFM variables are further standardised to ensure comparability, as they are measured on different scales.

The segmentation itself is performed using the K-means clustering algorithm. Clustering is used to uncover natural groupings of customers based on their RFM profiles, without predefined segment definitions. The Recency, Frequency and Monetary variables, after standardisation, are used as input variables, and customers are grouped into segments with similar behavioural characteristics. The number of clusters is determined using the Elbow Method, which evaluates the trade-off between model complexity and within-cluster variance. The resulting clusters are then interpreted in terms of customer behaviour and value. The clustering solution is further evaluated using internal validation metrics and visualisation techniques to assess its quality and interpretability.

The final step extends the analysis by introducing a temporal perspective. Customer activity is aggregated over time, primarily at the monthly level, in order to identify recurring patterns in purchasing behaviour. The analysis focuses on comparing how different customer segments behave across time, with particular attention to periods of increased activity. This allows the identification of segments that are more responsive to seasonal changes. The analysis remains descriptive in nature and focuses on identifying observable patterns rather than applying advanced time-series modelling techniques.

4 Analysis and Findings

This chapter presents the empirical analysis conducted on Mobilegear.cz transactional data and follows the CRISP-DM framework as a structured approach to the analytical process. Building on the methodological foundations, the chapter applies data-driven techniques to better understand customer behaviour, identify meaningful segments, and analyse seasonal purchasing patterns.

The chapter begins with the Business Understanding phase, which introduces the company context and defines the analytical objectives in relation to customer segmentation and seasonality. It is followed by the Data Understanding and Data Preparation phases, where the dataset is explored, cleaned, and transformed into a suitable format for analysis.

The Modelling phase then applies RFM analysis and K-means clustering to identify distinct customer groups based on purchasing behaviour. In addition, seasonality analysis is conducted to examine how customer activity changes over time and across segments. The results of these methods are subsequently assessed in the Evaluation phase, where their quality and interpretability are analysed.

Finally, the Deployment phase translates the analytical findings into practical business implications, focusing on marketing and inventory decision-making. Together, these steps provide a comprehensive view of customer behaviour and demonstrate how the CRISP-DM framework can be applied in a real-world e-commerce context.

4.1 Business Understanding

Mobilegear.cz is a Czech e-commerce company specialising in the sale of used and refurbished consumer electronics, with a primary focus on Apple devices such as iPhones, MacBooks, and related accessories. The company was founded around 2020 as a small startup with only a few employees and has since experienced steady growth. Over time, it has expanded its operations, moved to larger premises, and currently employs dozens of people. In addition to its online presence, Mobilegear also operates a physical retail store in Břeclav, Czech Republic.

The company's business model is based on sourcing used devices from both international suppliers and local buy-back programmes, followed by a structured internal process of testing, cleaning, and

classification. Each device undergoes functionality checks and is assigned a cosmetic grade before being offered for sale. This approach enables the company to resell used devices in a standardised and scalable way. Similar business models can be observed in other recommerce platforms such as Swappie or JabkoLevně, which operate in the same market segment.

As the company continues to grow and its customer base expands, understanding customer purchasing behaviour becomes increasingly important. In this context, data-driven approaches such as customer segmentation and the analysis of purchasing patterns can support more effective marketing activities, inventory planning, and overall business decision-making.

4.1.1 Objective of the Analysis

The objective of this analysis will be to examine customer behaviour in the context of Mobilegear.cz by combining customer segmentation techniques with a temporal perspective. The analysis will focus on identifying distinct customer groups based on their purchasing behaviour using RFM analysis and K-means clustering. This approach will make it possible to group customers according to how recently and frequently they purchase, as well as how much they spend, providing a structured view of differences in customer behaviour.

In addition to segmentation, the analysis will go beyond a static view by incorporating a seasonal perspective. The aim will be to explore how purchasing behaviour varies across customer segments over time and to identify recurring patterns in demand. By linking customer segments with temporal dynamics, the analysis will provide a more comprehensive understanding of how different types of customers behave throughout the year. These insights will be used to support practical implications, particularly in the areas of marketing activities and inventory planning.

4.1.2 Data Requirements

To achieve the defined objectives, it will be necessary to work with sales data, ideally sourced from a CRM system, together with customer-level information. Sales data should include transaction history, purchase frequency, and the monetary value of purchases. In addition, each transaction should contain time-related information, such as the date and time

of purchase. It is also important that individual customers can be consistently identified across transactions, to ensure accurate aggregation of their purchasing activity.

4.1.3 Data Risks and Ethical Considerations

When working with customer data, it is necessary to consider several risks related to data handling. Customer data often include sensitive information, such as email addresses or other personal identifiers, and therefore must be handled in accordance with applicable regulations, including GDPR. To minimise risks, personal data should be anonymised or otherwise protected to prevent identification of individual customers and avoid potential misuse or reputational damage.

At the same time, data quality represents a critical factor for the reliability of the analysis. For the results to be meaningful and applicable in practice, the data must be consistent, accurate, and sufficiently complete. Poor data quality may lead to biased or misleading outcomes, which could negatively affect subsequent business decisions. Therefore, ensuring reliable data and responsible, transparent handling of customer information is a key prerequisite for the analysis.

4.2 Data Understanding

This chapter focuses on the Data Understanding phase of the CRISP-DM framework. The aim of this step is to gain a clear overview of the dataset, assess its structure and quality, and identify key characteristics relevant for further analysis.

The chapter begins with a description of the data sources and their structure, followed by an overview of the dataset scope and basic statistics. Next, the quality of the data is evaluated, including completeness, consistency, and the presence of potential anomalies. Finally, exploratory data analysis is conducted to identify initial patterns in customer behaviour, such as purchase frequency, transaction values, and temporal dynamics.

These steps provide a foundation for the subsequent Data Preparation and Modelling phases, where the dataset is further processed and used for RFM analysis and clustering.

4.2.1 Data Sources and Structure

The dataset was provided by Mobilegear.cz and consists of B2C transactional data representing customer purchases on the company's e-commerce platform. The data covers the period from 1 January 2023 to 1 November 2025.

It is structured into two main tables. The first table represents order-level data (Orders), containing information about individual transactions, including order number, creation date, payment date, total order value, and a hashed customer email. The structure of this table is illustrated in Figure 1. The second table represents item-level data (Sales), where each row corresponds to a specific product included in an order. This table contains information such as product type, product name, item identifier, price, currency, and timestamp of the transaction, as shown in Figure 2.

Both tables are linked through the *order_number* variable, which allows the reconstruction of complete transactions and aggregation of item-level data to the order and customer level. The hashed email field serves as a unique identifier of customers, enabling the tracking of repeated purchases while preserving customer anonymity.

The data includes multiple product categories, such as iPhones, AirPods, MacBooks, iPads, and accessories. This provides a broader view of customer purchasing behaviour across different product types and supports further analysis of customer behaviour, including segmentation and seasonal patterns.

Figure 1: Orders table sample

order_number	created_at	e-mail hash	total	paid_at
11240144	2024-11-05T00:42:17+00:0	a5f1c5cd603e14fc3bbcd2c3f7ed4c7f7fed7ce91a9cd0c7945546b6693a333	14603.3	2024-11-05T17:01:12+00:00
11240148	2024-11-05T06:45:30+00:0	6ec7ff718e9bbf13e5e08659652b4d33b9b02167d6f631892ba19f26083742d	1998	2024-11-05T06:45:56+00:00
11240149	2024-11-05T06:52:12+00:0	94ccafe37aeb9cbeec1250f5c42cf85530985e1a1d9df8404a2941eb585a5e	4568	2024-11-05T06:53:30+00:00
11240151	2024-11-05T07:04:18+00:0	8049ffea15909040a4d6e9243a3d35635aad950f628c023cd01cf9d8a4b7b1	5247	2024-11-05T09:29:02+00:00
11240152	2024-11-05T07:21:39+00:0	235ae9375b92f1f6398a3c4cca074754c8cfc549a32df848e7d0b9bc04a796	18867	2024-11-05T09:24:48+00:00
11240154	2024-11-05T07:53:39+00:0	41276d889e14536fa99538169644d8306b2b6fe9c303310e2fcc7cbddcb7c8	968	2024-11-05T07:54:08+00:00
11240158	2024-11-05T09:35:06+00:0	78f0089ae27248e2ad75965f24e75bed00888dbcc491f607f5883de1d66858	11057	2024-11-07T12:45:20+00:00
11240160	2024-11-05T10:18:49+00:0	0e4cdd35518c19c39976fbbc5242d8098ea459c0e930b291c5923c570422	3099	2024-11-05T23:16:48+00:00
11240166	2024-11-05T12:13:33+00:0	7116317434191b0823eab6c2b365159cafa9074cf554ca4389e74376a46e8	12546	2024-11-05T12:14:23+00:00
11240169	2024-11-05T13:25:08+00:0	9b4632c43d7baa062c575e89dc86022835b8e2932d62f32d885699289724	8908	2024-11-05T13:37:37+00:00
11240170	2024-11-05T13:58:21+00:0	bf80135a7a59e99c5cedd46a8075ce9b1adb1ead6d4c9897374448368380e	1168	2024-11-06T06:10:45+00:00
11240173	2024-11-05T15:07:26+00:0	787eb2860f51db2eb12002371195fde442f48a1c31b4813e7704dedc66e8c5	7299	2024-11-05T17:09:24+00:00
11240176	2024-11-05T15:10:33+00:0	bf05604f04b0b13a938d06cf0673ab2f9bed8792d0a877744255296d7446fb2	4868	2024-11-05T15:11:07+00:00
11240178	2024-11-05T15:35:38+00:0	2f42086de901357b106e880e736486d7a3dadd3def6ab4cfc7ba9724c3468b	6068	2024-11-05T15:36:06+00:00
11240179	2024-11-05T16:28:09+00:0	dec052e7f2ca6dd76a2ea2c525824af7a04c7e6c11ff5e555fc286fbdd418f4b	11066.9	2024-11-05T16:30:46+00:00
11240180	2024-11-05T16:59:43+00:0	1bcce149c4945e350f4fcb16f7cf3f9c3078ceb506d9193ad800a67a592fcdc	17859	2024-11-05T16:59:59+00:00
11240181	2024-11-05T17:50:14+00:0	8045dd529bdfe64bac38b917821a34902b88abd49ab792980b68181b10ab5	8989.12	2024-11-05T17:51:30+00:00
11240182	2024-11-05T19:01:13+00:0	cce7bfc784c8a3792bdf8a38a2a1c8cb7fb0a54a2fb1cdbdd04a404f910f3a2	1908	2024-11-05T19:01:42+00:00
11240183	2024-11-05T19:03:54+00:0	d805ab8a668892aa15c51209c044379478fd1851b358bbc400a7d155ef354	2208	2024-11-05T19:05:31+00:00

Source: Author

Figure 2: Sales table sample

id	product_type	name	item_id	order_number	price	currency	ordered_at
20079	airpods	AirPods PRO 2 náhradní sluchátko levé (Lightning - A2699) - Bílá použité zařízení třídy Velmi dobrý+	211498	3250027	1699	CZK	2025-03-01T23:40:03+00:00
20093	airpods	AirPods PRO 1 s MagSafe nabíjecím pouzdrem - Bílá použité zařízení třídy Výborný	211484	3250024	3590	CZK	2025-03-01T19:01:49+00:00
20099	iphone	iPhone 13 PRO MAX 128GB Horsky modrá použité zařízení třídy Zánovní	211478	3250022	15390	CZK	2025-03-01T17:08:23+00:00
20102	iphone	iPhone 12 PRO MAX 128GB Tichomořsky modrá použité zařízení třídy Výborný	211474	3250021	11790	CZK	2025-03-01T16:30:04+00:00
20103	product_variant	Prémiový gelový magsafe obal pro iPhone 12 PRO MAX	211475	3250021	299	CZK	2025-03-01T16:30:04+00:00
20106	airpods	AirPods 4 (ANC) s bezdrátovým nabíjecím pouzdrem - Bílá použité zařízení třídy Výborný	211471	3250020	3890	CZK	2025-03-01T15:56:04+00:00
20112	airpods	AirPods 2 s Lightning nabíjecím pouzdrem - Bílá použité zařízení třídy Velmi dobrý	211457	3250017	1999	CZK	2025-03-01T14:22:52+00:00
20115	airpods	AirPods PRO 2 náhradní sluchátko pravé (Lightning - A2698) - Bílá použité zařízení třídy Rozbaleno	211450	3250015	1999	CZK	2025-03-01T13:30:54+00:00
20121	iphone	iPhone 13 mini 128GB Hvězdně bílá použité zařízení třídy Velmi dobrý+	211444	3250013	10390	CZK	2025-03-01T13:00:05+00:00
20124	airpods	AirPods 3 náhradní sluchátko levé (A2564) - Bílá použité zařízení třídy Velmi dobrý	211437	3250011	1099	CZK	2025-03-01T12:17:53+00:00
20130	iphone	iPhone 14 PLUS 512GB Žlutá použité zařízení třídy Rozbaleno	211430	3250009	15890	CZK	2025-03-01T12:00:06+00:00
20144	airpods	AirPods PRO 1 náhradní bezdrátové nabíjecí pouzdro (A2190) - Bílá použité zařízení třídy Velmi dobrý	211414	3250004	1999	CZK	2025-03-01T09:50:04+00:00
20150	airpods	AirPods 3 s Lightning nabíjecím pouzdrem - Bílá použité zařízení třídy Výborný	211408	3250002	2690	CZK	2025-03-01T08:20:03+00:00
20156	iphone	iPhone 14 PRO 256GB Vesmírně černá použité zařízení třídy Výborný	211399	2251206	17590	CZK	2025-02-28T19:40:07+00:00
20165	airpods	AirPods 3 náhradní sluchátko pravé (A2565) - Bílá použité zařízení třídy Nový	211393	2251204	1999	CZK	2025-02-28T18:27:27+00:00
20178	iphone	iPhone 13 PRO 1TB Grafitově šedá použité zařízení třídy Výborný	211386	2251202	15690	CZK	2025-02-28T18:20:09+00:00
20181	iphone	iPhone XS 64GB Stříbrná použité zařízení třídy Výborný	211383	2251201	4490	CZK	2025-02-28T18:40:07+00:00
20187	airpods	AirPods 3 s Lightning nabíjecím pouzdrem - Bílá použité zařízení třídy Rozbaleno	211376	2251199	2890	CZK	2025-02-28T16:10:07+00:00
20188	product_variant	Tenký gelový obal pro iPhone XR - Transparentní	211377	2251199	149	CZK	2025-02-28T16:10:07+00:00

Source: Author

4.2.2 Data Scope and Basic Statistics

The dataset consists of 24,171 unique customers, 31,247 orders, and 33,086 sales records. This indicates that, on average, each order contains slightly more than one item.

Out of the total number of customers, 5,197 have made two or more purchases. This corresponds to a repeat purchase rate of approximately 21.5%, indicating that a significant proportion of customers return to make additional purchases. At the same time, the majority of customers appear to make only a single transaction.

A more detailed view is provided by the distribution of orders per customer, shown in Figure 3. The results confirm that most customers have made only one purchase, while the number of customers decreases as the number of orders increases. Only a small fraction of customers exhibit higher purchase frequencies, forming a typical long-tail distribution.

Figure 3: Distribution of orders per customer

Orders per customer	Number of customers
1	18974
2	3954
3	828
4	311
5	86
6	8
7	3
8	1
9	1
10	1
18	1
92	1

Source: Author

From a structural perspective, the distribution of orders per customer is skewed towards lower frequencies, with a large group of one-time buyers and a smaller group of repeat customers. This imbalance is important for segmentation, as it influences the formation of clusters in the RFM analysis.

Overall, the size of the dataset can be considered sufficient for applying RFM analysis and K-means clustering, as it provides a large number of observations across customers and transactions. The repeat purchase rate is also adequate for identifying meaningful differences in customer behaviour.

However, the main limitation lies in the time span of the data, which covers less than three full years. This may restrict the ability to fully capture long-term seasonal patterns.

4.2.3 Data Quality Assessment

The dataset was examined to assess its overall quality and suitability for further analysis, focusing on completeness, consistency, integrity, currency, and the presence of outliers.

No missing or incomplete values were identified in the dataset, indicating a high level of completeness at the order level.

Data consistency was assessed through duplicate detection of order identifiers. A total of 14 duplicated order numbers were identified in the

dataset. This indicates minor inconsistencies in the uniqueness of transaction records; however, given the overall size of the dataset, this issue does not significantly affect its usability for further analysis.

Regarding time consistency, the analysis focuses primarily on the payment timestamp (*paid_at*), which reflects completed transactions. The dataset covers the period from 1 January 2023 to 1 November 2025, with an average transaction date of 27 July 2024. Quartile values indicate that 25% of transactions occurred before 30 November 2023, 50% by 12 August 2024, and 75% by 29 April 2025.

Currency consistency was also evaluated. The dataset is predominantly denominated in CZK (33,080 records), with only a negligible number of transactions in EUR (5 records).

Outlier analysis was conducted using descriptive statistics of the order-level monetary values. The average transaction value is 6,948.85 CZK, with values ranging from 0 to 81,293 CZK. Quartile analysis shows that 25% of transactions are below 2,068 CZK, 50% below 4,366 CZK, and 75% below 9,990 CZK. The presence of zero-value transactions was identified, which can be explained by items provided at no cost, such as complimentary accessories (e.g. protective glass) included with an order or promotional gifts. In total, 19 out of 31,246 transactions have a value of zero, indicating that such cases are rare and do not significantly affect the overall analysis. Higher-value observations correspond to typical prices of premium electronic devices and therefore do not indicate data errors.

Although the dataset shows a high level of completeness and consistency, several aspects should be considered when interpreting the results. The presence of duplicate order identifiers, although limited, may introduce minor inconsistencies in transaction-level analysis. Similarly, the existence of zero-value transactions and highly skewed monetary distributions reflects the specific nature of the recommerce market, but may also influence the distribution of RFM variables.

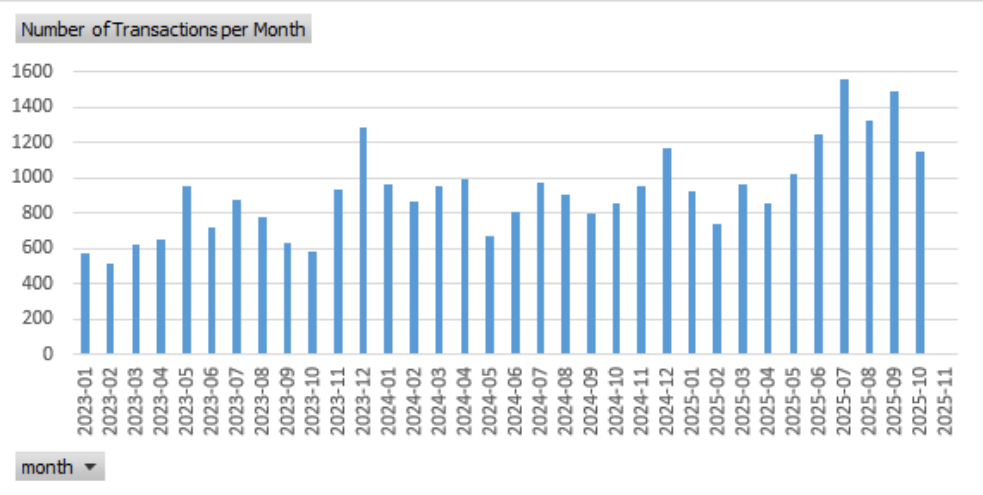
In addition, the strong skewness of purchase frequency and monetary value may affect the performance of clustering algorithms, as distance-based methods are sensitive to such distributions. Although these effects are partially mitigated in later preprocessing steps, they should be taken into account when interpreting the segmentation results.

4.2.4 Exploratory Data Analysis

As discussed in Subchapter 4.2.2, the dataset is characterised by a typical customer structure consisting of a large proportion of one-time buyers and a smaller group of repeat customers. This imbalance results in a long-tail distribution of purchase frequencies, which has important implications for subsequent segmentation.

Figure 3 illustrates the distribution of orders per customer and confirms that customer activity decreases as the number of purchases increases. This pattern highlights the heterogeneity of customer behaviour and provides a basis for applying RFM segmentation techniques.

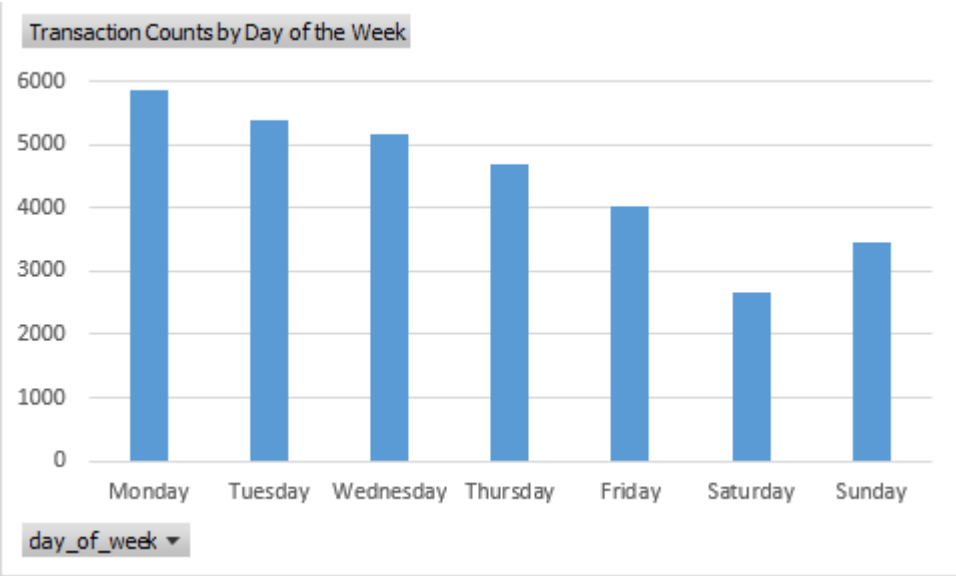
Figure 4: Number of Transactions per Month



Source: Author

The monthly distribution of transactions (Figure 4) indicates a relatively stable level of activity over time, with visible fluctuations across individual months. Several peaks can be observed, particularly towards the later part of the analysed period, where transaction volumes exceed earlier levels. This suggests a potential increase in customer activity or demand over time. At the same time, recurring variations across months point to the presence of seasonal patterns, although these are not uniformly consistent across all years.

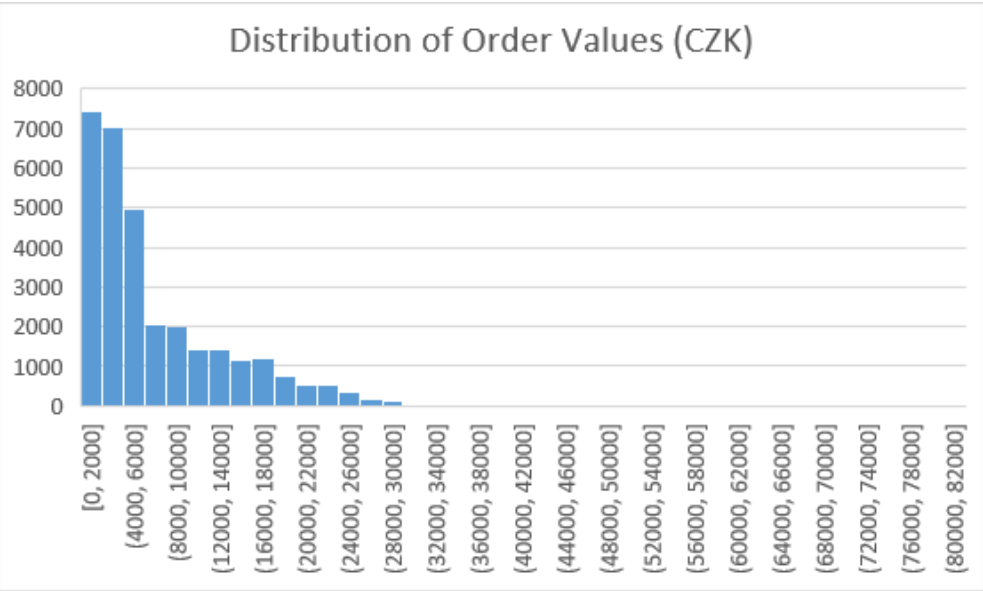
Figure 5: Transactions by Day of the Week



Source: Author

The distribution of transactions by day of the week (Figure 5) reveals a clear behavioural pattern. Transaction activity is highest at the beginning of the week, peaking on Monday, and gradually declines towards the weekend. The lowest number of transactions occurs on Saturday, while Sunday shows a slight increase but remains below weekday levels. This pattern indicates that customers are more active during working days, whereas weekend activity is comparatively lower.

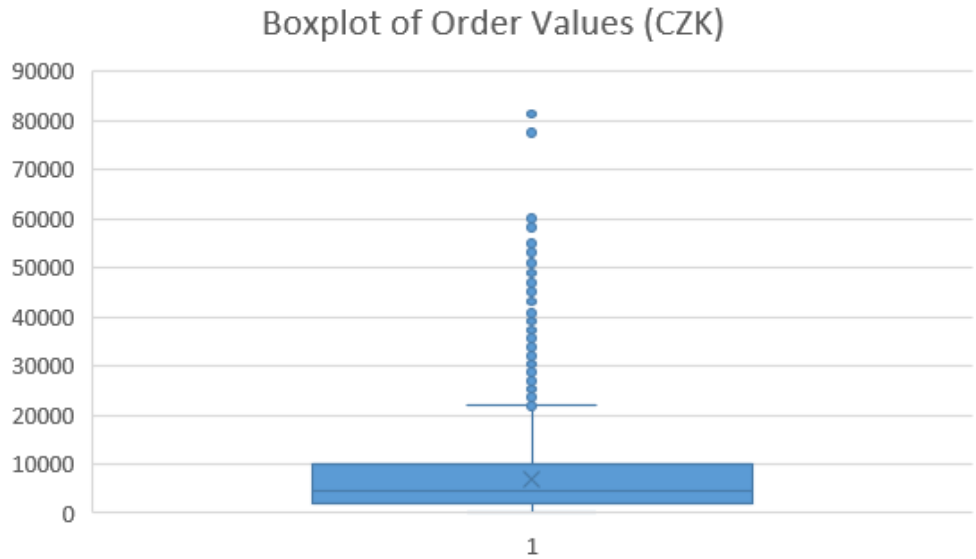
Figure 6: Distribution of Order Values (Histogram)



Source: Author

The distribution of order values (Figure 6) shows that the majority of transactions are concentrated in lower and mid-range price levels, with frequencies decreasing as the order value increases. This indicates that most purchases fall within relatively moderate price ranges, while high-value transactions occur less frequently.

Figure 7: Boxplot of Order Values



Source: Author

Further insights are provided by the boxplot of order values (Figure 7), which highlights the presence of a large number of higher-value observations above the upper quartile. The distribution is positively skewed, as confirmed by the skewness coefficient of 1.74. This indicates a long right tail, where a relatively small number of high-value transactions significantly exceed the typical order range.

These higher-value transactions correspond to purchases of premium electronic devices and are therefore expected within the context of the analysed market. At the same time, the concentration of most transactions in lower price ranges suggests a heterogeneous purchasing structure, which is relevant for subsequent RFM-based segmentation.

4.3 Data Preparation

This chapter describes the Data Preparation phase of the CRISP-DM framework, which follows the initial exploration of the dataset. The main objective of this step is to transform the raw data into a structured and analysis-ready form suitable for subsequent modelling.

The process begins with data selection and initial cleaning, where only relevant variables are retained and inconsistencies are addressed.

This is followed by data transformation, including adjustments of data types and time-related variables to enable accurate temporal analysis. Next, analytical datasets are constructed through aggregation and feature engineering, resulting in both customer-level and transaction-level datasets tailored to the objectives of the study. Finally, additional preprocessing steps are applied, including outlier detection and feature scaling, to improve the robustness of the clustering analysis.

These steps ensure that the dataset is consistent, comparable, and suitable for applying RFM analysis and K-means clustering in the following modelling phase.

4.3.1 Data Selection and Initial Cleaning

In the initial stage of data preparation, it was necessary to identify variables relevant for the RFM analysis and the seasonal analysis of customer behaviour. The original dataset contained a broader set of attributes that were not required for the objectives of this study.

For the calculation of RFM metrics, only key variables from the *orders* sheet were retained: *e-mail hash* as a customer identifier, *paid_at* as the transaction timestamp, and *total* as the monetary value of each order. The variable *order_number* was preserved as a linking key between the sheets.

The variable *created_at* was removed, as a single time indicator is sufficient for the analysis, with *paid_at* representing the actual completion time of the transaction.

From the *sales* sheet, only the variables *order_number* and *product_type* were retained. The product_type variable is relevant for the seasonal analysis, as it enables the examination of demand patterns across different product categories.

As observed in Section 4.2.3, the dataset contained a small number of transactions recorded in EUR (5 cases). These transactions were converted into CZK using a fixed exchange rate of 1 EUR = 24.40. The conversion was applied to the total variable to ensure consistency across the dataset.

Other variables, including *id*, *item_id*, *name*, *currency*, *ordered_at*, and *price*, were removed, as they were not relevant for the analysis. The total order value is already captured in the *total* variable in the *orders* sheet, making item-level prices unnecessary.

As mentioned in Section 4.2.3, a total of 14 duplicated order numbers (28 records) were identified in the dataset. A detailed inspection showed that each duplicated order number was associated with a different customer, different transaction value, and different purchased products. Given the very small number of such cases and the inconsistency of the records, it is assumed that these duplicates are the result of an internal system issue.

Therefore, these orders were not removed and were treated as unique transactions in the analysis. This approach is justified as the RFM analysis is based on the e-mail hash, which differs across these records and ensures correct customer-level aggregation.

These steps resulted in a reduced and cleaned dataset containing only variables necessary for further analysis.

4.3.2 Data Transformation

To enable time-based analysis, the *paid_at* variable was transformed from a string format into a standard datetime format. The original timestamps contained additional characters, including the *T* separator and timezone information. These elements were removed, and the timestamps were converted into a consistent datetime format suitable for further processing.

The timestamps were additionally adjusted from UTC to the local time zone (UTC+1) to better reflect actual customer behaviour patterns. This ensures more accurate interpretation of temporal features, particularly for analyses involving time-of-day and seasonal trends.

In addition, the total variable was checked and confirmed to be in a numeric format to ensure correct aggregation in subsequent analysis.

4.3.3 Dataset Construction and Feature Engineering

In this step, the prepared transactional data were transformed into analytical datasets suitable for further modelling and analysis. Two complementary datasets were created: a customer-level dataset for segmentation purposes and a transaction-level dataset for seasonal analysis.

First, the *orders* sheet was loaded, and the data were aggregated to the customer level using the e-mail hash. For each customer, key variables were calculated, including the *First_Purchase_Date* (minimum of *paid_at*) and *Last_Purchase_Date (maximum of paid_at)*, the total number

of transactions (*Frequency*), and total spending (*Monetary*). These variables form the basis of the RFM framework.

Additional variables were subsequently derived. The *Recency* variable was calculated as the number of days between the customer's last purchase (*Last_Purchase_Date*) and a reference date defined as the most recent transaction in the dataset. The *Average_Transaction_Value* was computed as the ratio of *Monetary* to *Frequency*. Furthermore, *Customer_Lifetime_Days* was defined as the number of days between *First_Purchase_Date* and *Last_Purchase_Date*.

The resulting customer-level dataset therefore consists of the following variables: *Customer_id*, *First_Purchase_Date*, *Last_Purchase_Date*, *Frequency*, *Monetary*, *Recency*, *Average_Transaction_Value*, and *Customer_Lifetime_Days*. The dataset was validated using descriptive statistics to ensure consistency and plausibility and was exported as *customer_data.csv* for further use in clustering analysis.

In parallel, a transaction-level dataset was constructed to analyse seasonal patterns. The *orders* sheet was merged with the *sales* sheet using the common key *order_number*, allowing the inclusion of product-level information, specifically the *product_type* variable.

From the transaction timestamp (*paid_at*), additional temporal variables were derived, namely *Month*, *Year*, and *Quarter*, which enable the identification of seasonal trends.

During the merging process, it was observed that not all transactions had corresponding product-level information. Specifically, 6,104 orders could not be matched with the *sales* sheet, resulting in missing values in the *product_type* variable. In contrast, 25,142 orders were successfully matched and enriched with product-level information. These unmatched observations were excluded from the dataset to ensure analytical consistency.

The final transaction-level dataset thus contains the variables *order_number*, *Customer_id*, *total*, *paid_at*, *product_type*, *Month*, *Year*, and *Quarter*, and serves as the basis for the subsequent seasonal analysis. The dataset was exported as *seasonal_data.csv*.

4.3.4 Outlier Detection and Feature Scaling

In order to improve the robustness of the following clustering analysis, additional preprocessing steps were applied to the customer-level

dataset. These steps focused on the identification and removal of extreme observations, followed by the standardisation of selected variables.

Outlier detection was performed using the Isolation Forest algorithm, which is designed to identify anomalous observations based on their relative isolation within the data space. The method was applied to the key behavioural variables *Recency*, *Frequency*, and *Monetary*. A contamination parameter of 2% was specified, reflecting the assumption that only a small proportion of customers exhibit atypical behaviour.

The results of the model indicate that 484 customers were classified as outliers, while 23,685 customers were identified as normal observations. The detected outliers correspond primarily to customers with unusually high transaction frequency or extreme monetary values, which could distort the clustering results. These observations were therefore removed from the dataset. As a result, the final dataset used for further analysis contains 23,685 customers.

Following the removal of outliers, feature scaling was applied to ensure comparability across variables. Since the RFM variables are measured on different scales, particularly the *Monetary* variable, which exhibits higher absolute values, standardisation is necessary to prevent any single variable from disproportionately influencing the clustering process.

The StandardScaler method was used to transform the variables *Recency*, *Frequency*, and *Monetary* to a standardised scale with a mean of zero and a standard deviation of one. The resulting dataset contains the variables *Customer_id*, *Recency*, *Frequency*, and *Monetary*, where the RFM variables are expressed in their standardised form.

This transformed dataset serves as the input for the modelling stage conducted in the following section.

While the removal of outliers improves the stability of the clustering results, it also represents a simplification of the data. Some extreme observations may correspond to genuinely high-value customers or atypical but valid purchasing behaviour. Their exclusion may therefore lead to a partial loss of information about the most valuable or unusual customer segments.

This trade-off between robustness and information loss should be considered when interpreting the clustering outcomes.

4.4 Modelling

This chapter presents the Modelling phase of the CRISP-DM framework, where the prepared data is used to apply analytical methods and generate meaningful insights. The aim of this step is to identify customer segments based on behavioural patterns and to analyse their purchasing dynamics over time.

The modelling process begins with RFM analysis, which transforms transactional data into a structured representation of customer value and activity. This is followed by K-means clustering, where customers are grouped into distinct segments based on their RFM characteristics. Finally, a seasonality analysis is conducted to examine temporal patterns in purchasing behaviour and to assess how these patterns differ across the identified customer segments.

Together, these steps provide a comprehensive analytical framework that combines behavioural segmentation with a time-based perspective, forming the basis for the interpretation of results in the following chapter.

4.4.1 RFM Analysis

In this step, the prepared customer-level dataset was used to perform RFM analysis.

The analysis was carried out on an aggregated dataset, where each customer was represented by three variables. These variables were created in the previous data preparation phase. *Recency* was calculated as the number of days since the last purchase, *Frequency* as the total number of orders, and *Monetary* value as the total amount spent by each customer.

To make the values comparable, each variable was transformed into a score from 1 to 5. This was done using a quintile-based approach. First, quantiles at the 20%, 40%, 60%, and 80% levels were calculated for each variable. Based on these thresholds, customers were divided into five groups.

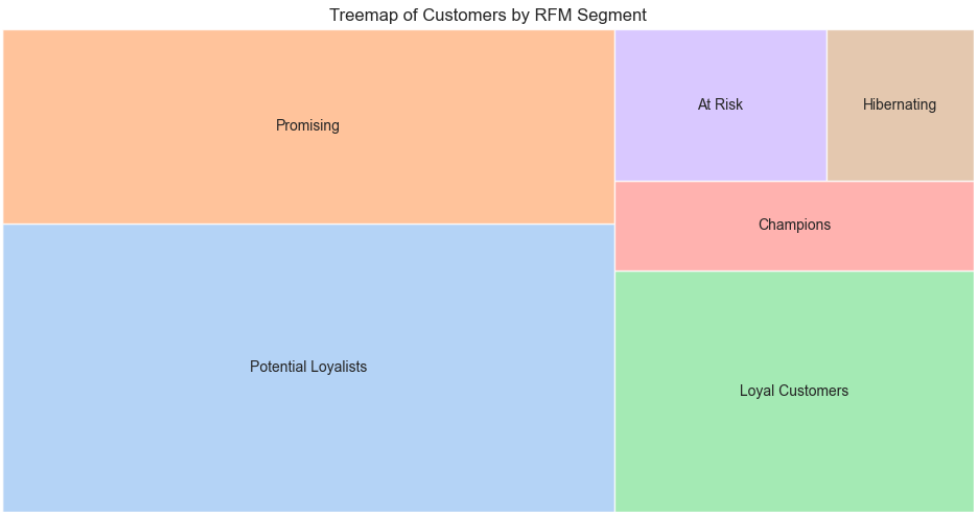
For *Recency*, lower values mean that the customer purchased more recently, which is considered better. Therefore, customers with the lowest *Recency* were assigned a score of 5, while those with the highest *Recency* received a score of 1. For *Frequency* and *Monetary* value, higher values are better. Customers with higher purchase frequency and higher

spending were assigned higher scores, while lower values received lower scores.

After assigning the individual scores, the three values were combined to describe the overall behaviour of each customer. First, the scores were joined into a three-digit RFM code (e.g., 555 or 115), which gives a more detailed view of the customer profile. Second, the scores were summed into a single value called the RFM combined score. This simplified the segmentation process.

The RFM combined score was then used to assign customers into predefined segments. These segments were based on score ranges and reflect typical customer behaviour groups used in marketing. Customers with the highest scores were labelled as "Champions", representing the most valuable and active customers. Slightly lower scores corresponded to "Loyal Customers", followed by "Potential Loyalists" and "Promising" customers. Customers with low scores were classified as "At Risk" or "Hibernating", indicating lower activity or decreasing engagement. The distribution of customers across these segments is illustrated in Figure 8.

Figure 8: Treemap: Number of Customers by Segment



Source: Author

As part of the analysis, several visualisations were created to support interpretation of the results. These included bar charts showing the number and percentage of customers in each segment, as well as plots comparing average Recency, Frequency, and Monetary values across segments. These visual outputs help to better understand the structure of

customer groups and make it easier to interpret differences between them.

Finally, the resulting RFM dataset was exported as *rfm_df.csv* and used as the input for the subsequent clustering analysis.

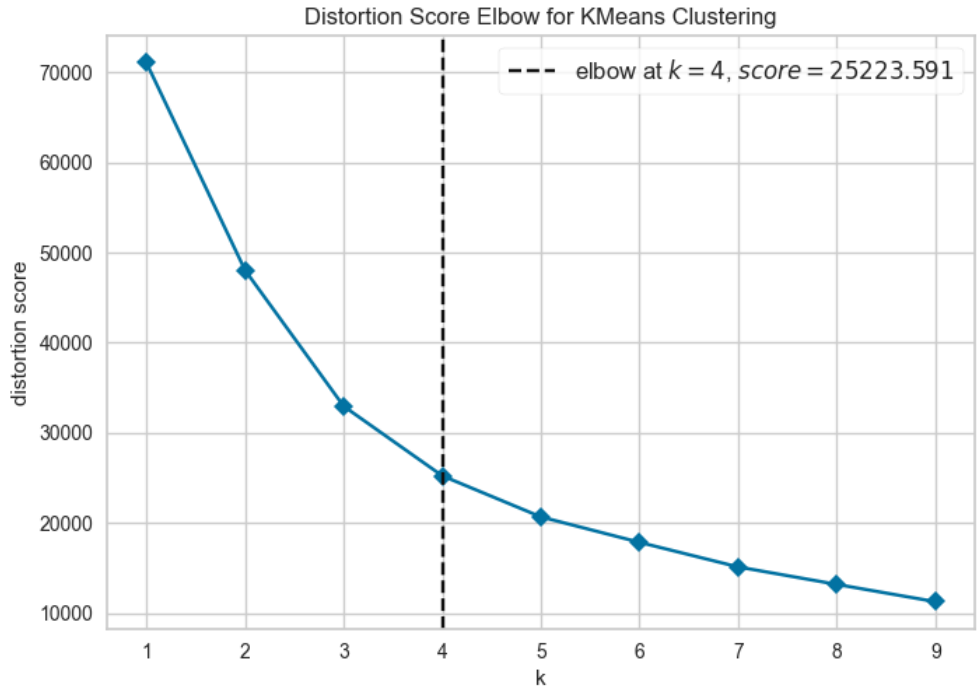
4.4.2 K-means Clustering

At this stage, K-means clustering was applied to the prepared customer-level dataset in order to identify groups of customers with similar purchasing behaviour. The clustering was performed using the *Recency*, *Frequency*, and *Monetary* variables, which had been previously standardised to ensure comparability.

First, the scaled RFM variables were selected from the dataset and used as the input for clustering. These variables were chosen because K-means is a distance-based algorithm and therefore requires variables to be on the same scale to avoid bias in the clustering process.

To determine an appropriate number of clusters, the elbow method was applied. The K-means algorithm was iteratively fitted for values of k ranging from 1 to 10, and the corresponding inertia values were recorded. Inertia represents the within-cluster sum of squared distances and is commonly used as a measure of cluster compactness. The results were visualised using the KElbowVisualizer, which allows for a clearer identification of the optimal number of clusters, as shown in Figure 9.

Figure 9: Elbow method for determining optimal number of clusters



Source: Author

The elbow method indicated a potential optimal number of clusters in the range of k = 3 to k = 4, as the reduction in inertia begins to stabilise after this point. While k = 3 could also be considered, k = 4 was selected for further analysis.

This decision was based not only on the elbow method, but also on the interpretability of the resulting clusters. The solution with four clusters provided a clearer and more meaningful segmentation of customers from a business perspective, allowing the identification of distinct groups such as high-value, repeat, new, and inactive customers.

This approach is consistent with the literature, which suggests that the selection of the number of clusters should not rely solely on a single metric, but should also consider the practical usability and interpretability of the segmentation results.

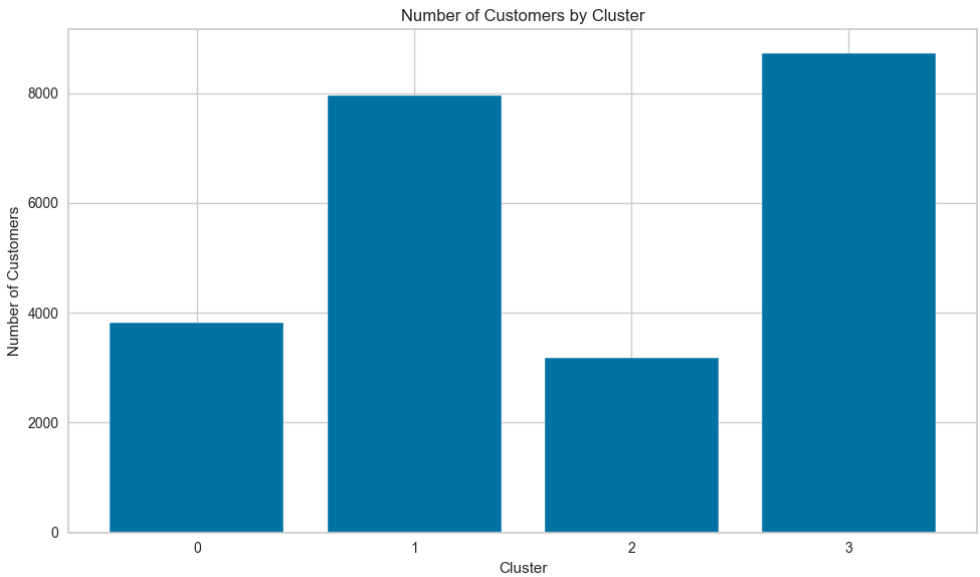
Based on this reasoning, the number of clusters was set to four. The K-means algorithm was then applied with this value, and each customer was assigned to one of the four clusters. The resulting cluster labels were stored in the dataset as a new variable called *Cluster*.

To better understand the structure of the clusters, the cluster centres were extracted. These represent the average scaled RFM values for

each cluster and provide an initial overview of the differences between customer groups.

In addition, the distribution of customers across clusters was analysed. The number of customers in each cluster was calculated and visualised using bar charts (Figure 10). A relative distribution in percentage terms was also computed to provide a clearer view of the overall structure of the segmentation.

Figure 10: Number of customers by cluster



Source: Author

After that, the cluster labels were merged back with the original customer-level dataset. This step allowed the clusters to be analysed using the original, non-scaled variables. For each cluster, average values of key variables were calculated, including *Recency*, *Frequency*, *Monetary*, *Average_Transaction_Value*, and *Customer_Lifetime_Days*. These aggregated statistics provide a basis for interpreting the characteristics of individual clusters in later stages of the analysis.

To improve interpretability, each cluster was further assigned a descriptive label based on its average RFM characteristics. These labels were not generated by the algorithm itself but were derived from the observed patterns in the data. Specifically, the clusters were labelled as *Inactive Customers*, *New Customers*, *Repeat Customers*, and *High-Value Customers*. The purpose of this step was to make the segmentation more

understandable from a business perspective and to support subsequent analysis and discussion.

Finally, the resulting dataset, containing both the cluster assignments and descriptive labels, was exported as *clustered_data.csv* and used for further evaluation and interpretation.

4.4.3 Seasonality Analysis

This part of the modelling focuses on the time dimension of customer purchasing behaviour. The aim is to identify recurring patterns in sales and to examine how these patterns differ across customer segments defined in the previous step.

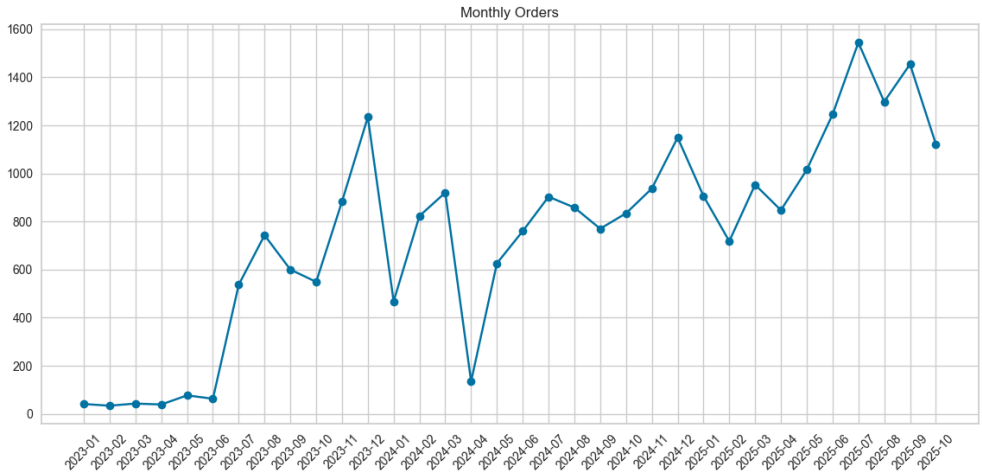
The analysis uses the transaction-level dataset enriched with cluster labels. This allows both overall seasonality and segment-specific behaviour to be analysed within the same framework.

The dataset was first prepared for time-based analysis. The transaction date (*paid_at*) was converted into a datetime format, and additional variables were created, including year, month, and a combined year-month variable. These variables were used to aggregate the data over time. In addition, specific seasonal periods were defined based on the business context. Three main periods were considered: Christmas (November–December), Back-to-School (August–September), and the remaining months labelled as regular periods.

To ensure accurate results, the analysis was performed at the order level. Duplicate item-level records were removed so that each order was counted only once. The last incomplete month (November 2025) was also excluded, as it contained only two orders and would therefore distort the results.

The first step examined overall seasonality. Monthly aggregates were calculated for the number of orders, total revenue, and number of customers. These values were visualised using time series plots to observe changes over time and identify periods of higher and lower activity, as illustrated in Figure 11.

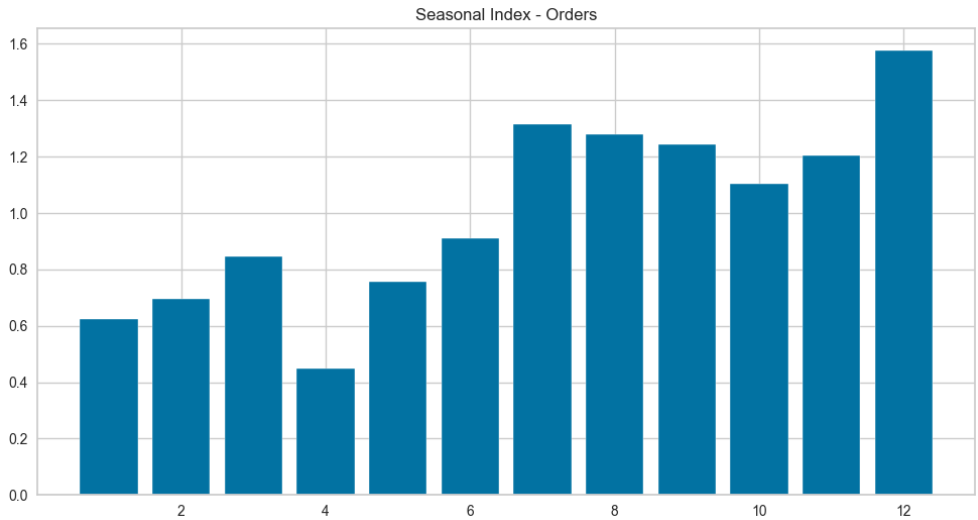
Figure 11: Monthly number of orders over time



Source: Author

To quantify these patterns, seasonal indices were calculated. The data were grouped by month and averaged across years to obtain typical monthly values. These averages were then compared to the overall mean. As a result, each month was assigned an index value indicating whether it performs above or below the average level, as shown in Figure 12.

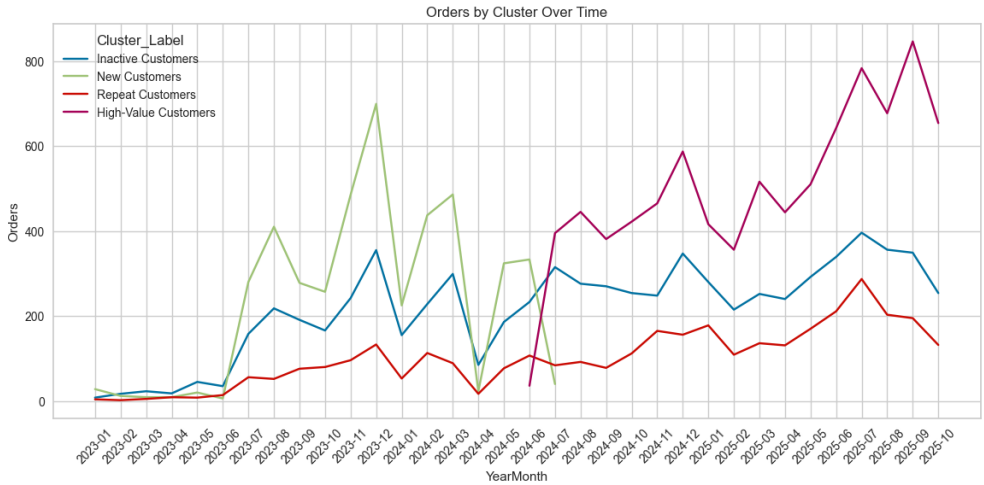
Figure 12: Seasonal index of orders by month



Source: Author

The analysis was then extended to customer segments. Monthly aggregates were computed separately for each cluster, allowing a comparison of how different groups behave over time, as illustrated in Figure 13. This makes it possible to identify whether certain segments are more sensitive to seasonal changes.

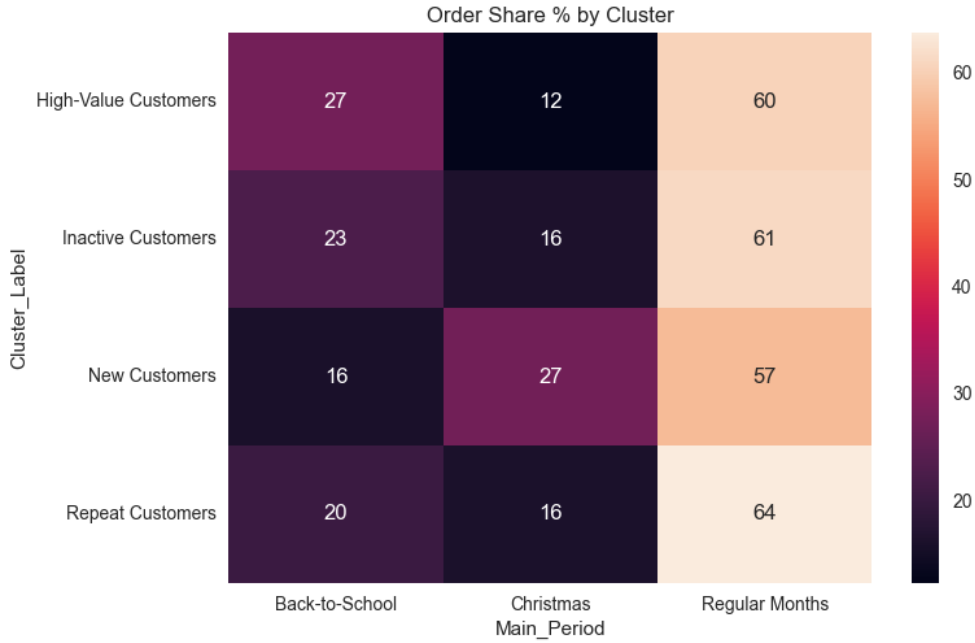
Figure 13: Monthly orders by cluster



Source: Author

Further insight was obtained by analysing the distribution of orders across the defined seasonal periods. For each cluster, the share of orders assigned to each period was calculated and is presented in Figure 14. This provides a simple measure of how strongly each segment is associated with specific parts of the year.

Figure 14: Share of orders by seasonal period and cluster



Source: Author

Additional comparisons were carried out for key periods, especially Christmas and Back-to-School. For each segment, the number of orders and revenue in these periods were compared with the rest of the year. The results were visualised using bar charts and heatmaps to support further analysis.

The outputs of this step provide a structured view of seasonal behaviour at both the overall and segment level. These results are used in the following section to interpret the findings and derive practical implications for marketing and inventory management.

4.5 Evaluation

This chapter presents the Evaluation phase of the CRISP-DM framework, where the results of the applied models are assessed and interpreted. The objective of this step is to evaluate the quality, validity, and practical relevance of the segmentation and analytical outputs obtained in the previous modelling phase.

The evaluation focuses on three main areas. First, the RFM segmentation is assessed in terms of the distribution of customers across segments and its implications for understanding customer value. Second, the K-means clustering results are evaluated using both internal validation metrics and cluster profiling to examine the robustness and interpretability of the identified groups. Finally, the seasonality analysis is evaluated to determine how purchasing behaviour varies over time across different customer segments. In addition to descriptive analysis, selected statistical tests are used to support the validity of the observed patterns.

These steps ensure that the results are not only methodologically sound but also meaningful from a business perspective, providing a solid basis for the final discussion and recommendations.

4.5.1 Evaluation of RFM Segmentation

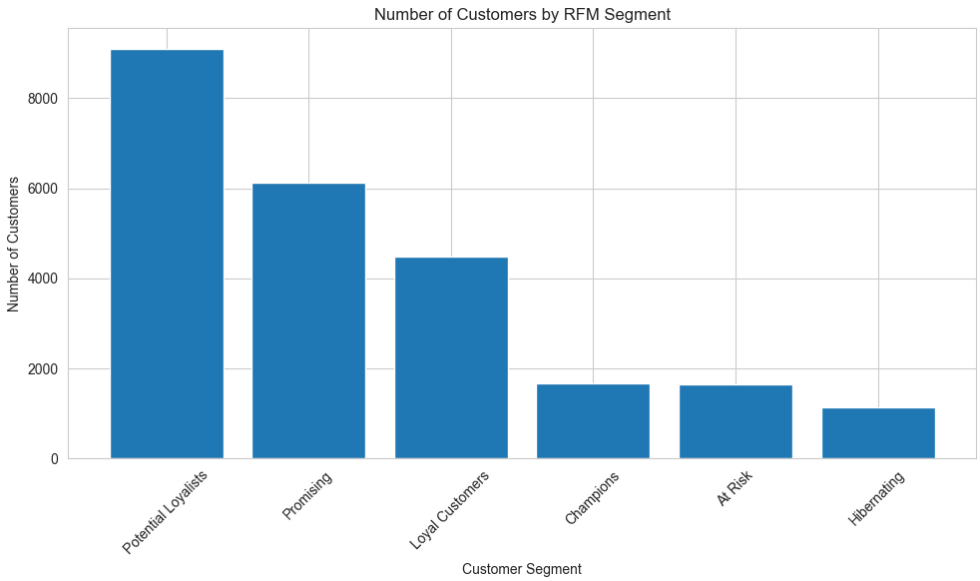
The distribution of customers across RFM segments provides an overview of the structure of the customer base. The results show that the largest segment consists of *Potential Loyalists* with 9,095 customers, followed by *Promising customers* with 6,130 customers and *Loyal Customers* with 4,476 customers. Smaller segments include *Champions* with 1,670 customers, *At Risk* with 1,648 customers, and *Hibernating customers* with 1,150 customers.

These results indicate that a large share of customers is concentrated in intermediate segments rather than in the most valuable or least active groups.

This distribution suggests opportunities for targeted marketing strategies, particularly focusing on converting *Potential Loyalists* and *Promising customers* into more valuable segments.

The distribution of customers across RFM segments is illustrated in Figure 15.

Figure 15: Number of Customers by RFM Segment



Source: Author

4.5.2 Evaluation of K-means Clustering

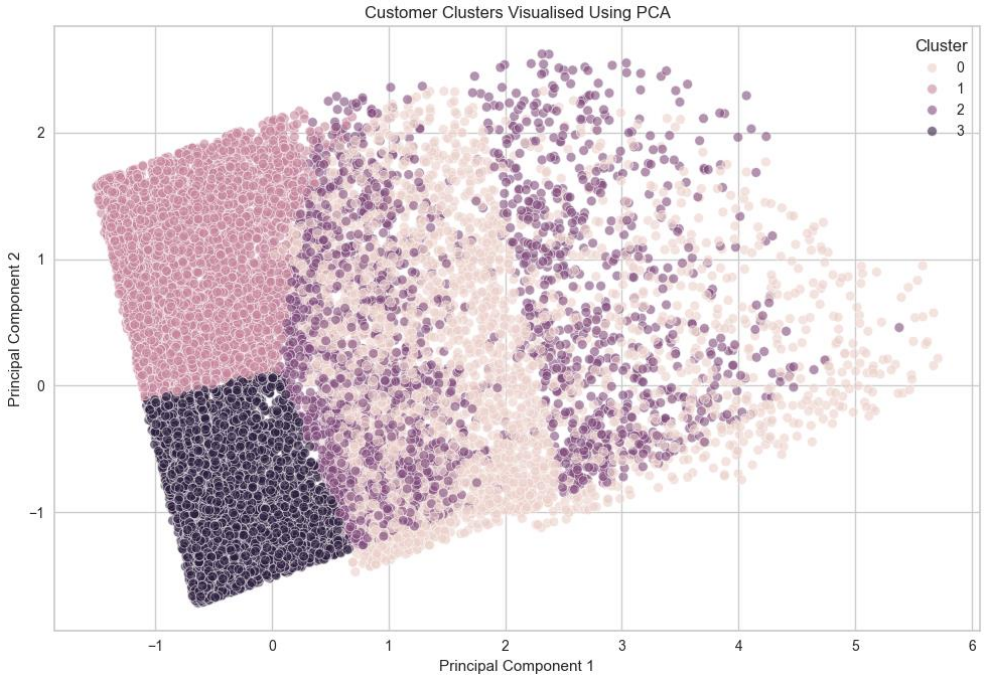
The distribution of customers across clusters shows that the largest cluster is Cluster 3 with 8,719 customers (36.8%), followed by Cluster 1 with 7,952 customers (33.6%). Smaller clusters include Cluster 0 with 3,828 customers (16.2%) and Cluster 2 with 3,186 customers (13.5%). This distribution suggests that most customers are concentrated in two dominant behavioural groups, while the remaining clusters represent more specific customer types. The cluster distribution was previously presented in Figure 10.

The quality of the clustering solution was evaluated using internal validation metrics. The silhouette score reached 0.414, indicating a reasonable degree of separation between clusters. The Calinski–Harabasz index was 14,342.84, which suggests strong cluster separation, while the Davies–Bouldin index reached 0.98, indicating acceptable cluster compactness. Taken together, these results suggest that the clustering solution is sufficiently robust for further interpretation.

To further assess the clustering structure, principal component analysis (PCA) was applied. The first two principal components explain approximately 78.3% of the total variance, specifically 45.8% and 32.6%. The PCA plot shows partial separation between clusters, although some

overlap is still visible, which is expected given the continuous nature of RFM variables. The PCA visualisation is presented in Figure 16.

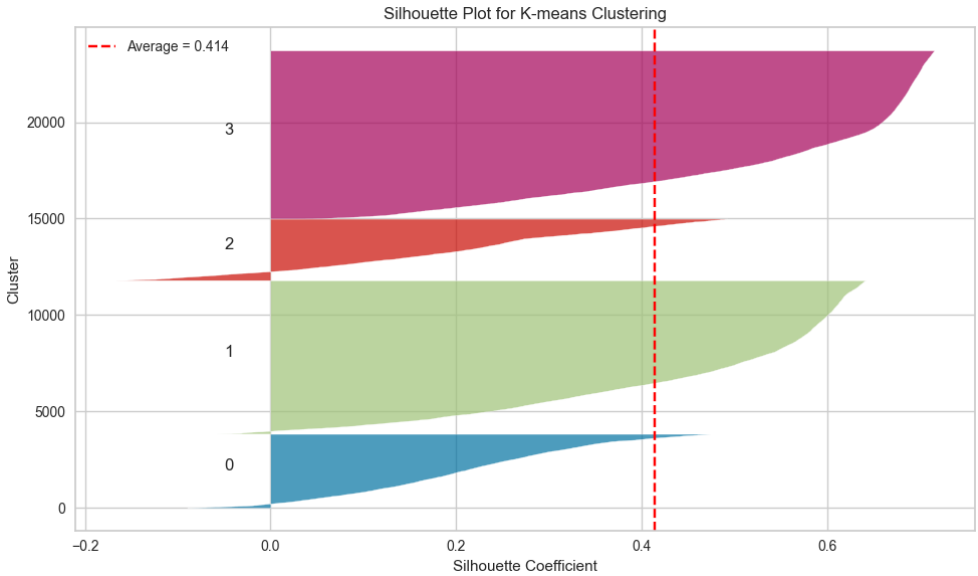
Figure 16: Customer Clusters Visualised Using PCA



Source: Author

The silhouette plot provides additional support for the selected clustering solution. Most observations have positive silhouette values, and the average silhouette score of 0.414 confirms a moderate but acceptable cluster structure. The silhouette plot is shown in Figure 17.

Figure 17: Silhouette Plot for K-means Clustering



Source: Author

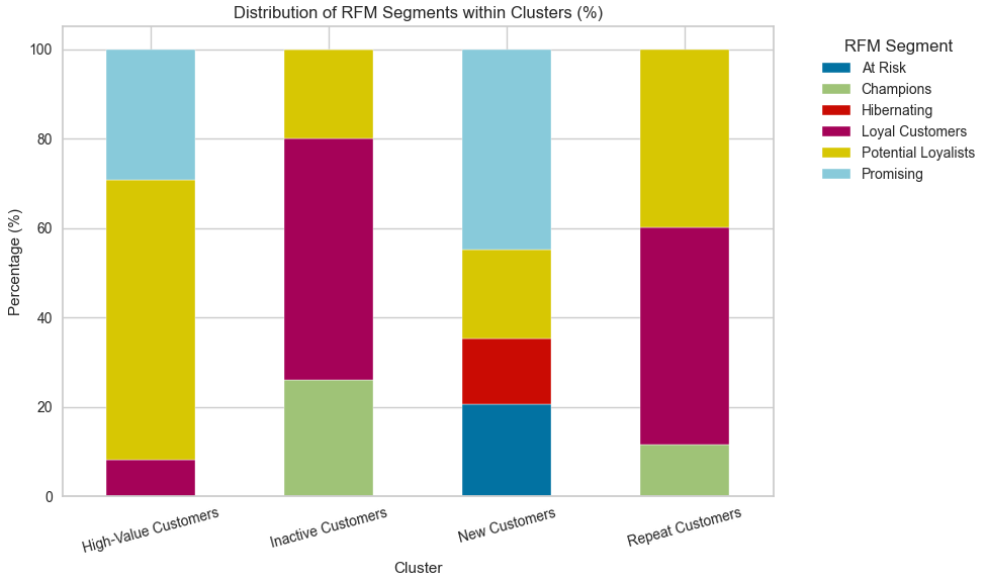
Cluster profiling was performed using the original non-scaled variables. The results show noticeable differences between clusters. *High-Value Customers* have the lowest average Recency at 216.0 days, but also the lowest Frequency at 1.00 orders and a relatively low Monetary value of 4,449.0. *Inactive Customers* show an average Recency of 390.4 days, Frequency of 2.26 orders, and Monetary value of 11,078.5. *New Customers* have the highest average Recency at 757.3 days, Frequency of 1.02 orders, and Monetary value of 5,581.6. *Repeat Customers* show the highest Monetary value at 22,379.8 and an average Frequency of 1.24 orders.

These results indicate that the assigned cluster labels should be interpreted as business-oriented summaries rather than exact descriptions of a single RFM dimension. In other words, the labels reflect overall behavioural patterns, but some of them do not correspond perfectly to intuitive expectations based only on Recency, Frequency, or Monetary value.

The relationship between clusters and RFM segments further supports the validity of the clustering approach. *High-Value Customers* are mainly composed of *Potential Loyalists* (62.53%) and *Promising customers* (29.34%). *Inactive Customers* are dominated by *Loyal Customers* (53.97%) and *Champions* (25.94%). *New Customers* consist primarily of

Promising (44.84%) and *At Risk* customers (20.72%), while *Repeat Customers* are mainly formed by *Loyal Customers* (48.68%) and *Potential Loyalists* (39.77%). This relationship is illustrated in Figure 18.

Figure 18: Distribution of RFM Segments within Clusters (%)



Source: Author

The observed relationship highlights the differences between rule-based RFM segmentation and data-driven clustering. While RFM segments are defined based on predefined scoring thresholds, K-means clustering groups customers according to overall similarity across the RFM variables.

As a result, clusters do not correspond exactly to individual RFM segments, but rather combine multiple segments with similar behavioural patterns. For example, the *High-Value Customers* cluster includes both *Potential Loyalists* and *Promising customers*, which suggests that these groups share similar overall behaviour despite differences in individual RFM scores.

This indicates that clustering captures broader behavioural structures that are not fully reflected in rule-based segmentation. At the same time, the partial overlap between clusters and RFM segments supports the consistency of the results and confirms that both approaches describe similar underlying customer patterns from different perspectives.

Overall, this comparison confirms that K-means clustering provides a meaningful extension to RFM segmentation by capturing patterns that are not fully represented by predefined scoring rules

4.5.3 Evaluation of Seasonality Across Clusters

The evaluation of seasonality focuses on differences in purchasing behaviour over time across customer clusters.

Cluster volatility analysis shows that *New Customers* have the highest variability in monthly order volume, with a coefficient of variation (CV) of 0.92, indicating strong fluctuations over time. This is followed by *Repeat Customers* with a CV of 0.69 and *Inactive Customers* with a CV of 0.55. In contrast, *High-Value Customers* show the lowest variability, with a CV of 0.37, which suggests more stable purchasing behaviour throughout the year. These results are summarised in Table 1.

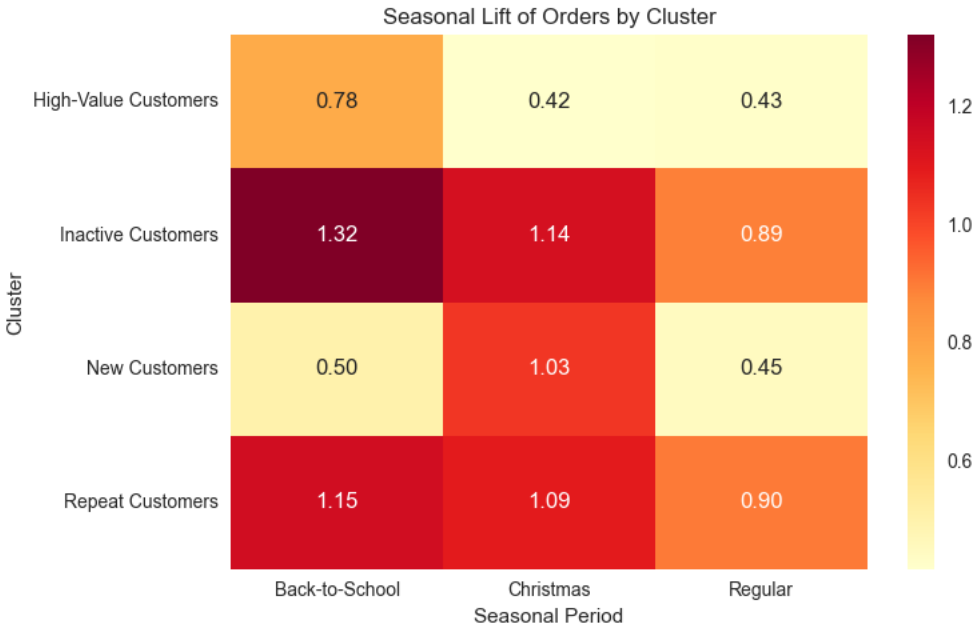
Table 1: Cluster Volatility Based on Monthly Orders

Cluster	Avg Monthly Orders	Std Monthly Orders	CV
New Customers	229.6	212.3	0.92
Repeat Customers	100.9	69.2	0.69
Inactive Customers	210.0	115.8	0.55
High-Value Customers	504.5	187.9	0.37

Source: Author

Seasonal lift analysis reveals how customer activity changes during key periods. *Inactive Customers* show the strongest seasonal effect, with a lift of 1.32 during Back-to-School and 1.14 during Christmas. *Repeat Customers* also show elevated activity, with lift values of 1.15 and 1.09, respectively. In contrast, *High-Value Customers* show lower activity during these periods, with lift values of 0.78 for Back-to-School and 0.42 for Christmas. *New Customers* show a mixed pattern, with a moderate increase during Christmas (1.03) but clearly lower activity during Back-to-School (0.50). These results are presented in Figure 19.

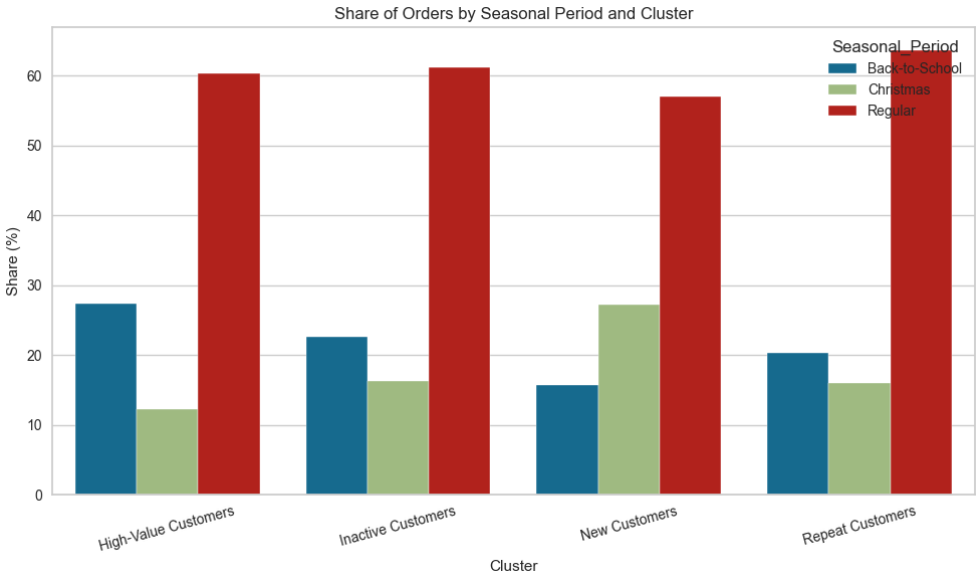
Figure 19: Seasonal Lift of Orders by Cluster



Source: Author

The distribution of orders across seasonal periods further supports these findings. *Inactive Customers* and *Repeat Customers* show a relatively stronger concentration of orders in seasonal periods, *while High-Value Customers* are more evenly distributed throughout the year. *New Customers* show a particularly strong share of orders during Christmas. This distribution is shown in Figure 20.

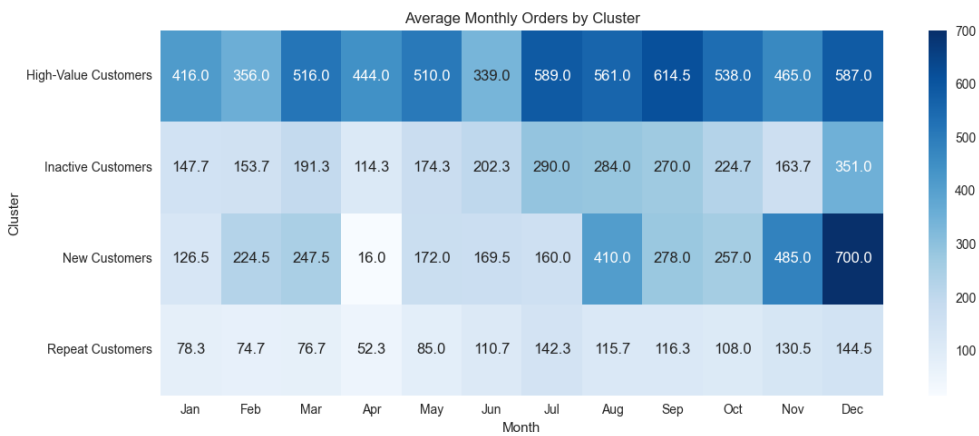
Figure 20: Share of Orders by Seasonal Period and Cluster



Source: Author

Finally, the monthly heatmap provides a more detailed view of temporal patterns. The results show that *New Customers* have a strong peak in December, while *Inactive Customers* also show increased activity towards the end of the year. In contrast, *High-Value Customers* maintain relatively stable monthly order volumes across the observed period. The heatmap is presented in Figure 21.

Figure 21: Monthly Orders by Cluster (Heatmap)



Source: Author

To complement the descriptive analysis, a chi-square test of independence was applied to examine whether the distribution of orders across seasonal periods differs significantly between customer clusters. The test was performed on the order-level dataset, where each order was counted only once. The chi-square test was selected because both variables are categorical and the objective was to assess whether cluster membership and seasonal purchasing period are statistically associated.

The results show a statistically significant association between customer cluster and seasonal period, $\chi^2(6) = 584.51$, $p < 0.001$. This indicates that the observed differences in seasonal purchasing behaviour across clusters are not random, but reflect systematic patterns in the data.

The comparison of observed and expected values further supports this conclusion. For example, *High-Value Customers* show fewer orders during Christmas than expected, while *New Customers* exhibit higher-than-expected activity in this period. Similarly, *Inactive Customers* display stronger than expected activity during Back-to-School. These deviations confirm that different customer segments respond differently to seasonal periods.

Overall, this statistical test supports the findings from the descriptive analysis. Seasonal patterns are therefore not only visually observable, but also statistically significant across customer clusters.

4.6 Deployment

This chapter addresses the Deployment phase of the CRISP-DM framework, where the results of the analysis are translated into practical business applications. The objective of this step is to demonstrate how the developed models and insights can support real-world decision-making.

The chapter begins by interpreting the analytical findings in a business context, focusing on their implications for marketing and inventory management. It then outlines how the results can be integrated into existing business processes, including the use of CRM systems and basic analytical tools. Furthermore, specific marketing applications are proposed for individual customer segments, reflecting differences in their behaviour and value. Finally, the chapter discusses monitoring and maintenance of the models to ensure their long-term relevance.

These steps show how the analytical outputs can be effectively utilised in practice, even without complex technical implementation, and highlight the role of data-driven approaches in supporting business strategy.

4.6.1 Translation of Results into Business Practice

The deployment phase focuses on translating analytical results into practical business use. In this thesis, the deployment is not implemented technically. Instead, the findings are interpreted and used to propose recommendations for marketing and inventory management in the context of Mobilegear.cz.

The results provide a structured view of customer behaviour. RFM segmentation indicates that most customers are concentrated in intermediate segments, mainly Potential Loyalists and Promising customers. This suggests that a large part of the customer base has the potential to increase its value. Smaller segments, such as Champions or At Risk customers, represent more specific groups that require a different approach.

Clustering further simplifies this structure into four main groups: High-Value Customers, Repeat Customers, Inactive Customers and New Customers. These clusters capture overall behavioural patterns and make the results easier to apply in practice. The findings confirm that customer value is unevenly distributed and that individual groups differ in their purchasing behaviour.

Adding a temporal perspective provides additional insight. Some groups, especially Inactive Customers and Repeat Customers, show higher activity during key periods such as Back-to-School and Christmas. In contrast, High-Value Customers behave more consistently throughout the year. New Customers show a strong increase in activity during Christmas, which highlights the importance of this period for customer acquisition.

These insights can support business decisions. From a marketing perspective, the results highlight the importance of targeting specific customer groups. Intermediate segments represent an opportunity to increase customer value through targeted campaigns, while Inactive Customers can be addressed through reactivation efforts during peak periods.

From an inventory perspective, the findings help identify periods of higher demand. Increased activity during specific months suggests the

need for higher stock levels in advance. More stable segments allow for more predictable demand planning.

Combining customer segmentation with seasonal analysis provides a clear basis for decision-making. It links customer behaviour with time patterns and supports more effective marketing and inventory decisions.

4.6.2 Integration into Business Processes

The analytical results can be integrated into existing business processes in a relatively simple way. Since Mobilegear.cz already uses a CRM system and email marketing tools, the identified customer segments can be linked to individual customer records. CRM systems provide a central platform for managing customer information and support the use of segmentation for targeted communication and customer management (Wilson et al., 2024).

Customers can be assigned to segments based on the RFM analysis and clustering results. This assignment does not need to be updated in real time. A regular update, for example on a monthly basis, is sufficient to reflect changes in behaviour. The segmentation process can be implemented using standard tools such as Excel or simple scripts, and the results can be imported into the CRM system.

Once the segments are available in the CRM, they can support basic customer management and communication. Email campaigns can be targeted to specific groups, allowing different types of messages to be sent based on customer behaviour, without the need for complex automation. This approach is consistent with the use of data-driven marketing strategies, where segmentation enables more personalised and effective customer engagement (Hu & Basiglio, 2023).

Seasonal insights can also be included in planning processes. Marketing activities can be scheduled according to expected peaks in demand, while inventory decisions can reflect anticipated increases in customer activity. These adjustments can be based on simple reports and historical data.

The integration of the results does not require major changes to existing systems. Segmentation can function as an additional layer of information. Even a basic implementation can improve decision-making by providing a clearer view of customer behaviour over time.

4.6.3 Marketing Applications by Segment

The identified customer segments can support more targeted marketing activities. Each segment shows different purchasing behaviour and therefore requires a different approach.

High-Value Customers represent a stable and important group. The main objective is retention. This can be supported through personalised communication, early access to selected products, or occasional loyalty rewards. Given their consistent behaviour, frequent discounting is not necessary.

Repeat Customers show relatively high spending but less stable behaviour. The focus for this group is to increase purchase frequency. This can be achieved through targeted product recommendations, limited-time offers, or reminders based on previous purchases.

New Customers represent an entry-level group with limited purchase history. The main objective is to encourage a second purchase. Follow-up emails, introductory offers, or personalised product suggestions can support this goal. Their strong activity during the Christmas period suggests that onboarding communication should be particularly active after this season.

Inactive Customers show lower engagement but respond more strongly during seasonal periods. This makes them suitable for reactivation campaigns. Time-limited discounts, seasonal promotions, or reminder emails can help bring these customers back.

Overall, segmentation allows marketing activities to be better aligned with customer behaviour. Even simple adjustments in communication and timing can improve campaign effectiveness.

4.6.4 Monitoring and Model Maintenance

To ensure that the results remain relevant over time, it is important to regularly monitor customer behaviour and update the segmentation. Customer preferences may change, which can reduce the accuracy of the model if it is not updated.

Several key performance indicators (KPIs) can be used to track performance. These include customer retention rate, average order value, purchase frequency, and the distribution of customers across segments. The use of KPIs allows organisations to evaluate the effectiveness of segmentation-based strategies and adjust their actions accordingly (Chowdhury, 2024).

The segmentation can be updated on a regular basis, for example monthly or quarterly. This involves recalculating RFM metrics and reassigning customers to segments. Over time, customer behaviour may shift, and segmentation models should be treated as dynamic rather than static tools (Quach et al., 2025).

Regular monitoring and periodic updates help maintain the usefulness of the segmentation and ensure that it continues to support business decision-making.

5 Discussion and Conclusions

This chapter interprets the results of the analysis and places them in a broader theoretical and practical context. Building on the findings obtained in the previous chapter, it evaluates their implications for understanding customer behaviour and their relevance for business decision-making.

The chapter begins with a discussion of the main results, linking them to existing literature and highlighting their contribution to customer segmentation and seasonality analysis. It then outlines the managerial implications, focusing on how the findings can support marketing activities and inventory planning in practice. Next, the limitations of the study are addressed, including constraints related to the dataset, methodology, and scope of the analysis. Finally, the chapter concludes by summarising the key contributions of the thesis and suggesting directions for future research.

Together, this chapter provides a comprehensive reflection on the analytical outcomes and their significance within the context of the recommerce market and data-driven customer analysis.

5.1 Discussion

This thesis analysed customer behaviour in the Mobilegear.cz dataset by combining RFM segmentation, K-means clustering, and a seasonal perspective. The results show that most customers are concentrated in intermediate RFM segments, particularly Potential Loyalists and Promising customers. This suggests that a large part of the customer base has growth potential but has not yet reached the highest-value categories. At the same time, smaller segments such as Champions or At Risk customers represent more specific behavioural patterns that require a different approach.

The findings confirm that customer value is not evenly distributed. A relatively small group of customers generates higher value, while the majority shows lower purchase frequency and lower spending. This type of long-tail distribution is typical for e-commerce environments and reflects the heterogeneity of customer behaviour. The results therefore support the idea that behavioural segmentation provides a more meaningful basis for understanding customers and for designing targeted

strategies than simple descriptive characteristics (Duhita Wahyu Maulida and Dachyar, 2023; Abbasimehr & Shabani, 2019).

The RFM model proved to be a suitable tool for this analysis. It allows transactional data to be transformed into an interpretable structure of customer value and behaviour. This is consistent with existing research, which highlights RFM as a simple and effective method for behaviour-based segmentation in CRM (Yeh et al., 2009; Khumaidi et al., 2023). At the same time, the analysis reflects some of the known limitations of the model. RFM captures customer behaviour only through transactional variables and does not fully represent long-term value or a broader behavioural context.

The clustering step further simplified the segmentation structure and grouped customers into four main categories. These clusters provide a more practical view of customer behaviour and make the results easier to interpret. The clusters are not fully separated, which reflects the continuous nature of customer behaviour and the strong presence of low-frequency customers in the dataset. This is consistent with typical e-commerce patterns, where a large proportion of customers make only a single purchase.

An important contribution of this thesis is the inclusion of a temporal perspective. The results show that customer activity varies over time and that this variation differs between segments. Inactive and Repeat Customers show stronger responses during seasonal periods such as Back-to-School and Christmas, while High-Value Customers behave more consistently. This finding is important because it addresses one of the key limitations of classical RFM, which is its static nature and limited ability to capture behavioural changes over time (Abbasimehr & Shabani, 2019).

The seasonal patterns also reflect the specific context of the recommerce market. Demand is influenced not only by general shopping cycles but also by factors such as product replacement, price sensitivity, and external events. This suggests that customer behaviour is shaped by both segment characteristics and time-related factors. As a result, segmentation becomes more informative when it is combined with a temporal perspective.

From a methodological point of view, the combination of RFM analysis and K-means clustering proved to be appropriate for the available data. The approach is relatively simple and can be applied without complex data requirements. At the same time, the results are influenced by

the structure of the dataset, especially the high proportion of one-time buyers and the skewed distribution of purchase frequency. These characteristics limit the precision of segmentation but do not reduce its practical usefulness.

Overall, the analysis suggests that the selected approach was suitable for the objectives of the thesis. It enabled the identification of meaningful customer groups, revealed differences in purchasing behaviour, and showed that seasonal effects vary across segments. The study therefore contributes by extending traditional segmentation with a temporal dimension and by providing a more comprehensive understanding of customer behaviour in a recommerce environment.

5.2 Managerial Implications

The results of this thesis provide several practical implications for business decision-making in the recommerce e-commerce context. The combination of RFM segmentation, K-means clustering, and seasonal analysis makes it possible to better understand customer behaviour and to translate these insights into concrete actions.

From a segmentation perspective, the results show clear differences between customer groups. Intermediate segments, such as *Potential Loyalists* and *Promising customers*, represent a key opportunity for increasing customer value. These customers usually have low purchase frequency but show potential for further engagement. They can be targeted with follow-up communication shortly after their first purchase, for example through personalised email recommendations or small incentives that encourage a second transaction. The goal is to increase purchase frequency and gradually move customers into more valuable segments.

High-Value Customers require a different approach. This group shows relatively stable behaviour over time and is less affected by seasonal changes. For this reason, retention should be prioritised over discount campaigns. Suitable actions include personalised communication, early access to selected products, or exclusive offers that support long-term relationships without reducing margins.

Inactive Customers represent another important segment. These customers purchased in the past but have not been active recently. Reactivation campaigns can be used, for example through targeted emails or

time-limited discounts. These campaigns can be triggered after a defined period of inactivity.

New Customers form a large part of the customer base and typically make only one purchase. For this segment, onboarding is important. This can include post-purchase communication, cross-selling of complementary products such as accessories, and gradual engagement through CRM campaigns.

The seasonal analysis adds another important dimension. Customer activity changes over time, and some segments respond more strongly during specific periods. *Inactive Customers* and *Repeat Customers* show higher activity during Back-to-School and Christmas periods. This suggests that marketing activities for these segments should be increased during these times. In contrast, *High-Value Customers* show lower sensitivity to seasonal changes, which means that seasonal campaigns may be less effective for this group.

From an inventory perspective, the results provide useful input for demand planning. Higher demand during specific periods, especially in late summer and at the end of the year, suggests that stock levels should be increased in advance. At the same time, more stable segments allow for more predictable planning.

Finally, the results can be used in CRM systems to support automated decision-making. Customer segments can be stored and used to trigger targeted campaigns based on behaviour and timing. This makes marketing communication more personalised and efficient.

5.3 Limitations and Future Research

Despite these contributions, several limitations should be considered when interpreting the results. The first limitation relates to the dataset. Although the number of customers and transactions is sufficient for segmentation, the time span of the data is relatively short and does not cover multiple full cycles of long-term seasonal patterns. This may limit the ability to fully capture recurring trends.

An additional limitation relates to data preprocessing decisions. The removal of outliers and the transformation of variables, although necessary for the application of clustering methods, may affect the resulting segmentation. In particular, extreme but valid customer behaviour may

be underrepresented in the final model, which can influence the interpretation of high-value segments.

Another limitation is the structure of customer behaviour. A large proportion of customers made only a single purchase, resulting in a skewed distribution of purchase frequency. This affects the segmentation process and reduces the ability to distinguish between different types of repeat behaviour.

The analysis is also limited by the type of available data. The dataset contains only transactional information and does not include demographic variables, behavioural engagement data, or customer interaction history. This restricts the depth of segmentation and the ability to explain customer behaviour beyond purchase activity.

From a methodological perspective, the use of the classical RFM model has its own limitations. The model focuses on past transactions and does not capture long-term customer value or future behaviour. It also does not incorporate temporal dynamics directly, which required an additional seasonal analysis in this study.

Future research could address these limitations in several ways. One possible direction is the use of extended models, such as CLV-based segmentation or enhanced RFM variants that include additional behavioural or temporal variables. Another option is the application of alternative clustering techniques that may better capture complex patterns in skewed data.

Further research could also benefit from a longer time horizon and the inclusion of additional data sources, such as customer demographics, browsing behaviour, or marketing interactions. Finally, the practical impact of segmentation could be evaluated through experimental approaches, such as A/B testing of targeted campaigns, to assess its effect on customer engagement and business performance.

5.4 Conclusion

This thesis demonstrated that combining RFM segmentation with K-means clustering is an effective approach for analysing customer behaviour in an e-commerce environment. The analysis identified distinct customer segments that differ in purchasing behaviour and value, providing a structured understanding of the customer base.

The results revealed clear differences in customer behaviour across segments and highlighted opportunities for further development. The inclusion of a seasonal perspective showed that customer activity varies over time and that these variations differ between segments.

The findings have practical relevance for business decision-making. They provide a basis for more targeted marketing activities, improved timing of campaigns, and better inventory planning. The study therefore supports the use of data-driven approaches in managing customer relationships and demand.

In summary, the thesis contributes to the field of customer segmentation by extending the traditional RFM approach with a temporal dimension. This combination provides a more comprehensive view of customer behaviour and offers practical insights for the recommerce context. Future research can build on this approach by incorporating additional data and more advanced analytical methods.

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Appendix A Python Analysis Code

A.1 Python Notebook

The Python code used in this thesis is available in the following GitHub repository:
https://github.com/premop614/customer-segmentation-and-seasonality-analysis

The repository contains the scripts used for RFM analysis, K-means clustering, and data visualisation.